On-Manifold Strategies for Reactive Dynamical System Modulation With Nonconvex Obstacles

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***Abstract*—In this work, we present a novel, reactive, modulated control strategy based on dynamical systems (DS) for planning in the context of multiple nonconvex obstacles. Our DS modulation strategy leverages an on-manifold planning methodology and provides several methods for real-time on-manifold navigation around nonconvex obstacles. We introduce a sample-based obstacle representation for complex, nonconvex obstacles, as well as a projection-based method for representing surfaces, such as tables, cylinders, and ellipsoids. These representations can be combined to represent multiple obstacles and obstacle types (sample- or projection-based) with a single, continuously differentiable function. We validate our approach in several real-world scenarios, including navigation within (simulated) constrained environments, as well as reactive control of a real 7-DoF manipulator with dynamic obstacles (including humans) while utilizing a 1 kHz control loop rate. Using our sample-based representation, we can calculate the obstacle representation function in less than 1 ms with up to 35 k points using a CPU implementation, and up to 600 k points with a GPU implementation.**

***Index Terms*—Concave obstacles, dynamic obstacle avoidance, dynamical systems (DS), real-time control and motion planning, safety constraints, safety guarantees, reactive motion planning.**

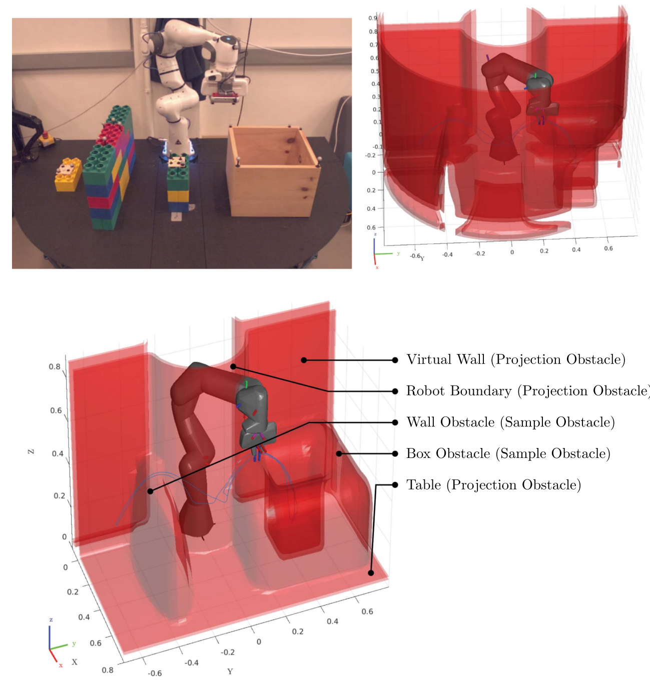

Fig. 1. Proposed modulation approach applied to the end-effector of a robot manipulator as a real-time, reactive, motion planner. Our method is able to navigate around nonconvex obstacles with a control frequency of 1 kHz, while updating the obstacle states at 100 Hz.

I. Introduction

STATE-dependent dynamical systems (DS) offer adaptivity, reactivity, and robustness to perturbations, uncertainties, and changes in dynamic environments for robot motion planning [16], [27], [30], [32]. One particular appeal of such methods is their ability to operate at high bandwidths, enabling safe navigation within a variety of contexts. However, even the most advanced state-of-the-art DS-based obstacle avoidance techniques rely upon the assumption that obstacles can be represented with predefined geometric shapes, such as spheres, cylinders, ellipsoids, polygonal, or star-shaped objects [6], [7], [15], [18], [23], [33]. Further, when considering multiple dynamic obstacles, these methods either resort to an averaging of multiple (per-obstacle) modulations [15], [18], [33], a composition of multiple (per-obstacle) repulsive/potential functions [6], [7], [23], or, more recently, a composition of multiple obstacles into one star-shaped obstacle [15]. To avoid undesired behaviors (e.g., local minima) such combinations must be carefully designed to guarantee impenetrability (the robot will never penetrate an obstacle's boundary) and convergence (the robot will converge to a single target/attractor). While many objects and surfaces can be represented by convex or star-shaped obstacles, such representations may become too conservative when considering constrained or dense environments. In the latter case, this can pose a practical limitation, as existing methodologies do not preserve the properties of the individual modulations when the obstacles are combined to form nonconvex obstacles (excluding star-shapes). In this work, we consider all obstacles that can be represented as (possibly disconnected) smooth manifolds,

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including multiple obstacles, geometric obstacles, and convex or nonconvex[1] sample-based geometries. Existing DS-based methods that do consider nongeometric concave obstacle representations (i.e., sensor-based representations, such as point clouds or depth data) suffer from the introduction of a potentially infinite number of spurious attractors (undesired local minima on the obstacle boundary) [18], [33].

We offer a DS-based approach to avoiding dynamic complex, nonconvex obstacles in constrained workspaces. Our method guarantees impenetrability, preserves stability, and can ensure convergence in the presence of multiple nonconvex obstacles.[2] We introduce a novel DS methodology that can accommodate nonconvex obstacles, as well as two novel obstacle representation functions capable of modeling complex obstacles (see Fig. 2) and workspaces while representing them as a single object boundary function. The contributions of this article are as follows.

1) A novel DS-based modulation strategy for nonconvex obstacles that utilizes manifold navigation methods based on approximated geodesics of the extended isosurface of the nonconvex obstacle representation—see Section V.
2) Two novel obstacle representations, including a computationally efficient sample-based representation for describing arbitrary nonconvex obstacles, and projection-based representation for describing continuous, well-defined surfaces, such as planes, columns, and ellipsoids—see Sections VI-A and VI-B, respectively.

To the best of the authors' knowledge, this is the first demonstration of a reactive control approach that can operate in the presence of nonconvex obstacles while preserving guarantees of stability and convergence. Furthermore, empirical evaluations demonstrate real-time ($<$1 ms) reactive collision avoidance and target tracking on both 2-D and 3-D examples, as well as on a physical robot with a human collaborator and a series of obstacles (for an illustrative example, see Fig. 1).

The rest of this article is organized as follows. In Section II, we cover work related to reactive motion planning, with a focus on existing DS-based modulation approaches. In Section III, we define our problem formulation and objectives and provide preliminaries of the DS-based obstacle avoidance approach we build upon in Section IV. We introduce our novel on-manifold modulation strategies in Section V and present the proposed obstacle representations in Section VI. Finally, we validate our presented contributions via simulated and real-world robotic experiments with increasing levels of complexity in Section VII and offer a thorough discussion of our results and the limitations of this work in Section VIII. Finally, Section IX concludes this article.

II. Related Work

The works we have drawn for comparison highlight local or reactive approaches to motion planning, with a focus on existing research in DS-based motion planning. We also discuss additional approaches, such as fast replanning methods based on model predictive control (MPC).

Local or reactive approaches attempt to modify the dynamics of a system when in proximity to an obstacle. Early work in this area included the development of artificial potential fields [19], where obstacles are modeled as exerting artificial forces that, while causing a system to avoid the obstacle, could also induce local minima and artificial stopping points in a robot's path. Navigation functions [30] resolved this issue and introduced mappings that enabled representation of star-shaped obstacles; harmonic potential functions [10] extended this to fields with a single minima, but remain difficult to construct for nongeometrically parameterizable obstacles or environments. Artificial potential fields were further extended to operate with depth cloud information [13], as well as to modify existing trajectories, such as in the elastic strips framework [4]. Vector field approaches have a similar premise, and are utilized to generate static, cell-based vector fields [23], or in an "almost global feedback" motion planning framework [28] wherein vector fields around obstacles modify local dynamics.

Inspired by harmonic potential fields [10], a DS modulation approach was introduced [18] that could ensure convergence to the target, but was limited to spherical obstacle representations. This was later extended to handle convex and concave obstacles from point cloud data [33], but this approach could induce an infinite number of spurious targets. The modulation approach [18] was, then, expanded upon to handle obstacles represented as ellipsoids, concave star-shaped, and polygonal volumes with convergence guaranteed via contraction theory [15]. However, the method has a single failure case where the obstacle can still be penetrated. Reactive, sensor-based methodologies that do not use modulation have also been effective for real-time obstacle avoidance. For example, Tulbure and Khatib [37] used global elastic planning with local artificial potential fields to dynamically navigate around obstacles in real-time, using point-cloud data and a pipeline that effectively handles partial occlusions. Shahriari et al. [36] demonstrated a lightweight collision avoidance strategy based on a blending of reference direction and collision avoidance vectors—an approach verified through Lyapunov stability theory. Roelofsen et al. [31] demonstrated collision avoidance using a velocity obstacle approach while incorporating sensing limitations such as a limited field of view. These methods, while effective at ensuring obstacle avoidance, do not encompass a wide range of possible sensing modalaties, and do not provide theoretical guarantees demonstrating convergence to the target position, or they are limited to 2-D cases.

One alternative to the aforementioned methods based on potential functions or DS-modulation is to formulate a constrained optimization problem [2], [17], [26], [35], [38]. Constrained optimization techniques, however, can scale poorly with the number and complexity of nonconvex constraints, which can become impractical for real-time deployment, where such optimization techniques may fail to provide a solution within the requisite time frame without careful initialization [26]. When considering spherical obstacle representations, the complexity of the constraints is drastically reduced; enabling near real-time

[1]Nonconvexity, in this setting, refers to any obstacle containing at least one concave region on the surface.

[2]Depending upon the obstacle escape strategy employed, and under certain conditions; see, Section V-C

MPC approaches to create collision free paths on-the-fly [17]. A recent sampling-based method solved the path integral optimal control problem [38], and can achieve real-time adaptation by utilizing GPU-based parallel computation [2]; a similar methodology has also been employed [35] for 2-D collision avoidance. When making use of parallel computation, these methods can calculate plans at frequencies of up to 100 Hz, yet, they rely heavily on GPU acceleration and assume parameteric obstacle representations.

Recently, several works have investigated the encoding of such collision constraints as metrics (rather than forces or potentials) from a differential geometry perspective [6], [7], [24]. Specifically, Riemannian metrics have been used to deform the manifold representing the robot (or task) configuration space in order to prevent constraints from being violated by stretching the space infinitely in the direction of the constraint [7], [24]. By relying upon the curvature of the manifold rather than potential or repeller functions, one can avoid introducing spurious attractors; these metrics, however, are difficult to design and whether they are applicable to arbitrary nonconvex obstacles has yet to be shown. Further, a predefined notion of the underlying task or obstacle manifold is necessary, limiting the approaches to well-known, predefined manifolds.

III. Problem Statement

Let $\mathbf{x} \in \mathbb{R}^d$ represent the d-dimensional state-space of a robotic system. In this work, we consider the state-space to be the Cartesian (task) space of the end-effector — i.e., $d > 2$. We assume the robot's motion can be guided by a first-order, autonomous DS, $f_g(\mathbf{x}(t))$, describing the nominal motion plan for a robot starting from outside the obstacle boundary, such that

$$\dot{\mathbf{x}}(t) = f_g(\mathbf{x}(t)), \quad \lim_{t\to\infty} \mathbf{x}(t) = \mathbf{x}^* \tag{1}$$

$f_g(\cdot) : \mathbb{R}^d \to \mathbb{R}^d$ is a continuous and differentiable vector-valued function representing a DS (vector field) that converges to a single stable equilibrium point $\mathbf{x}^* \in \mathbb{R}^d$—i.e., an attractor.

In this setup, we assume a holonomic robot is operating within a constrained workspace in proximity to potentially dynamic obstacles. The workspace limits and moving objects are modeled as obstacles the robot should avoid, and are described by closed regions $\mathcal{O} \subset \mathbb{R}^d$, with $\partial\mathcal{O}$ representing the boundary of a given region. Since the nominal motion generated by (1), referred to as global DS, does not consider obstacle or workspace constraints, we follow the approach from Zadeh and Billard [18] and Huber et al. [15] to ensure collision avoidance.

Formally, we propose a DS modulation strategy that *reshapes* the flow of the global DS w.r.t. a set of K complex, nonconvex obstacles $\mathcal{O} = \{\mathcal{O}^k\}_{k=1}^K$, such that

$$\dot{\mathbf{x}}(t) = M(\mathcal{O}, \mathbf{x}(t)) \cdot f_g(\mathbf{x}(t)). \tag{2}$$

Nominally, $M(\mathcal{O}, \mathbf{x}(t)) \in \mathbb{R}^{d\times d}$ is an auxiliary, continuous, matrix-valued function parameterized by the obstacles $\mathcal{O}$. Equation (2) is referred to as the *modulated dynamics* and is required to satisfy the following criteria.

1) *Impenetrability:* The generated dynamics will never penetrate any obstacle boundary from set $\partial\mathcal{O} = \{\partial\mathcal{O}^k\}_{k=1}^K$.
2) *Exit strategies:* The generated dynamics will always move away from the obstacle boundaries $\partial\mathcal{O}$, even in enclosed concave regions.
3) *Convergence:* Any motion starting outside obstacle boundaries $\partial\mathcal{O}$ reaches the attractor $\mathbf{x}^*$.
4) *Computational Efficiency:* The technique should operate at high frequency in order to guarantee strict human-safety requirements.

Notations: For readability, we omit the time-dependency of $\mathbf{x}(t)$, and parameter $\mathcal{O}$ in $M(\mathcal{O}, \mathbf{x})$, from all further equations and references to the state-space and modulation matrix M.

IV. Preliminaries

In this section, we provide the theory behind DS-based motion planning for obstacle avoidance and obstacle representation, beginning with a detailed discussion of how DS are constructed, and how modulations can generate appropriate behavior while in proximity to obstacles. We also provide an analysis of the limitations of current approaches that motivated our proposed approach.

A. DS Modulation

A global DS, $f_g(\mathbf{x})$, can be reshaped (or modulated) to generate new dynamics by premultiplying it with an auxiliary continuous matrix-valued function $M(\mathbf{x}) \in \mathbb{R}^{d\times d}$. If $M(\mathbf{x})$ is full-rank $\forall \mathbf{x} \in \mathbb{R}^d$ and locally active in a compact set that does not include the attractor, $\chi \subset \mathbb{R}^d \setminus \mathbf{x}^*$, it will preserve the attractor of the global DS, $f_g(\mathbf{x}(t))$, and inherits the asymptotic stability properties of the global DS [3], [21].

As introduced by Zadeh and Billard [18] and later extended in work by Huber et al. [15], by constructing $M(\mathbf{x})$ using a continuously differentiable scalar-valued function representing the distance to the obstacle boundaries $\partial\mathcal{O}$, we can avoid penetrating $\mathcal{O}$. The result is a collision avoidance behavior that emulates harmonic potential field approaches; i.e., impenetrability and no local minima [10], [20]. However, depending upon the construction strategy of $M(\mathbf{x})$ and the type of obstacles being considered (convex, star-shaped, or concave) the stability and convergence properties may not be preserved, due to the rank of $M(\mathbf{x})$ diminishing.

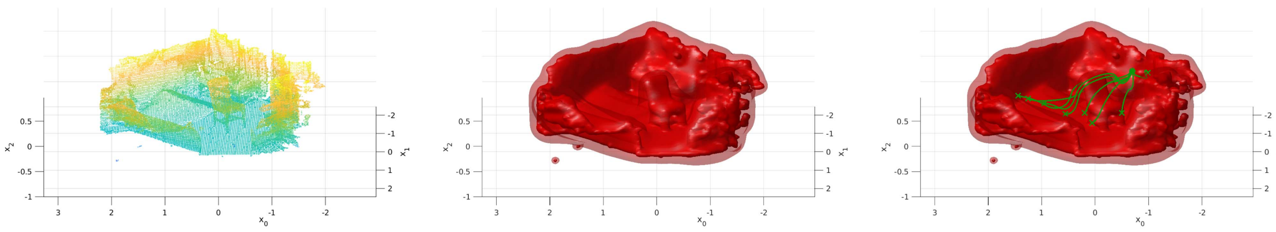
B. Obstacle Representation

In order to enable obstacle avoidance using the aforementioned modulation strategy, we assume the obstacle, $\mathcal{O}$, is known, and that an explicit representation of its shape exists through a scalar-valued function $\Gamma(\mathbf{x}) : \mathbb{R}^d \to \mathbb{R}$. $\Gamma(\mathbf{x})$ describes the proximity of the robot state to the obstacle boundary (with $\Gamma(\mathbf{x}) = 1$ at $\partial\mathcal{O}$) and is constructed such that

$$\Gamma(\mathbf{x}) \leq 1 \iff \mathbf{x} \in \mathcal{O} \quad \text{(Inside Obstacle)} \tag{3}$$

$$\Gamma(\mathbf{x}) = 1 \iff \mathbf{x} \in \partial\mathcal{O} \quad \text{(At Boundary)} \tag{4}$$

$$\begin{gathered}\forall \alpha > \beta, \mathcal{O}_\alpha \supset \mathcal{O}_\beta, \quad \text{(Monotonic)} \\ \mathcal{O}_\alpha = \{\mathbf{x} | \Gamma(\mathbf{x}) \leq \alpha\}, \mathcal{O}_\beta = \{\mathbf{x} | \Gamma(\mathbf{x}) \leq \beta\}.\end{gathered} \tag{5}$$



(a) (b) (c)

Fig. 2. Proposed modulation approach enables robust, real-time navigation around complex isosurfaces such as those from pointclouds. The pointcloud with $\sim$43 k points shown in this figure is from the Redwood dataset [8], and the obstacle exit strategy employed is presented in Section V-C2 with each modulation step taking < 15 ms with a CPU implementation. (a) Input: Pointcloud. (b) Isosurfaces: $\Gamma(\mathbf{x}) = \{1.01, 1.4\}$. (c) Planning: RK4 simulations.

For any given obstacle $\mathcal{O}$, $\Gamma(\mathbf{x})$ should be continuously differentiable ($\mathcal{C}^2$ differentiability) for all $\mathbf{x} \in \mathbb{R}^d$, with derivative $\nabla\Gamma(\mathbf{x})$ subject to $\|\nabla\Gamma(\mathbf{x})\| > 0 \forall \mathbf{x} \notin \mathcal{O}$. Further, in (5), we define a strict monotonicity requirement for nonconvex obstacles—a generalization of the original approach [18]. This implies that the set of all points $\mathbf{x} \in \mathbb{R}^d$ within the boundary of $\Gamma(\mathbf{x}) = \alpha$ subsumes regions with lower gamma values [i.e., that $\Gamma(\mathbf{x}) = \alpha$ is a continuous isosurface containing all isosurfaces of lower $\Gamma(\mathbf{x})$ values]. For spherical obstacles, this can be stated as follows:

$$\exists \mathbf{r} \text{ s.t. } \quad \forall t_1 \geq t_2 \geq 0 \quad \forall \mathbf{u}, \Gamma(\mathbf{r} + t_1\mathbf{u}) \geq \Gamma(\mathbf{r} + t_2\mathbf{u}). \quad (6)$$

Note that in the case of multiple obstacles, it is assumed that the free space around the obstacles is connected.

C. DS Modulation for Obstacle Avoidance

To avoid obstacles in the path of motion described by $f_g(\mathbf{x})$, a modulation matrix can be calculated as follows [18]:

$$M(\mathbf{x}) = H(\mathbf{x})\Lambda(\mathbf{x})H(\mathbf{x})^T. \quad (7)$$

$M(\mathbf{x})$ is analogous to an orthogonal eigendecomposition, composed of an orthonormal basis, $H(\mathbf{x})$, and diagonal scaling matrix, $\Lambda(\mathbf{x})$. The orthonormal basis, $H(\mathbf{x})$, is aligned with the gradient and tangent space of the obstacle representation function, $\Gamma(\mathbf{x})$, using the gradient direction, $\hat{\mathbf{n}} = \nabla/\Gamma(\mathbf{x})\|\nabla\Gamma(\mathbf{x})\|$. The basis is defined as follows:

$$H(\mathbf{x}) = \begin{bmatrix} \hat{\mathbf{n}} & \mathbf{e}_1 & \ldots & \mathbf{e}_{d-1} \end{bmatrix}, \quad H(\mathbf{x})^T H(\mathbf{x}) = I \quad (8)$$

where $\mathbf{e}_1, \ldots, \mathbf{e}_{d-1}$ represent orthogonal basis vectors to $\nabla\Gamma(\mathbf{x})$, which are utilized in the calculation of the modulation.[3] The basis representation is illustrated in Fig. 3. $\Lambda(\mathbf{x})$ describes the modulation itself, redistributing velocity based on proximity to the obstacle

$$\Lambda(\mathbf{x}) = \text{diag}\left(\begin{bmatrix} 1 - \frac{1}{\Gamma(\mathbf{x})} & 1 + \frac{1}{\Gamma(\mathbf{x})} & \ldots & 1 + \frac{1}{\Gamma(\mathbf{x})} \end{bmatrix}\right). \quad (9)$$

Note, this formulation causes large values of $\Gamma(\mathbf{x})$ to result in a modulation that approximates the identity matrix (i.e., $\lim_{\Gamma(\mathbf{x})\to\infty} \Lambda(\mathbf{x}) = \mathbb{I}_{d\times d}$). For small values of $\Gamma(\mathbf{x})$, the modulation acts to reduce velocity in the direction of $\nabla\Gamma(\mathbf{x})$ and increase velocity in the tangent plane. Velocity in the direction of the gradient only approaches zero when $\Gamma(\mathbf{x}) = 1$. For notational convenience, we refer to the individual elements of $\Lambda(\mathbf{x})$ as λ_i, such that $\Lambda(\mathbf{x}) = \text{diag}([\lambda_1 \quad \lambda_2 \quad \ldots \quad \lambda_d])$.

[3]Note that for $d = 3$, the basis can be described by $\hat{\mathbf{n}} = \mathbf{e}_1 \times \mathbf{e}_2$, whereas for $d = 2$, $\mathbf{e}_1 = R_{\pi/2}\hat{\mathbf{n}}$, where $R_{\pi/2}$ is the $\pi/2$ rotation matrix. In general, an orthogonal basis can be determined using QR factorization with $\nabla\Gamma(\mathbf{x})$ as the first vector, and a full-rank matrix (such as the identity matrix). However, as shown in Section IV-C3, an explicit basis does not need to be calculated.

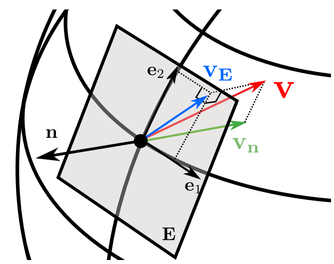


Fig. 3. Illustration of vector decomposition using orthogonal basis $H(\mathbf{x}) = [\mathbf{n} \quad \mathbf{e}_1 \quad \mathbf{e}_2]$. Here, $\mathbf{v_n}$ is the component of $\mathbf{v}$ along axis $\mathbf{n} = \nabla\Gamma(\mathbf{x})$, whereas $\mathbf{v_E}$ is the component within the tangent plane, defined by $\mathbf{E} = [\mathbf{e}_1 \quad \mathbf{e}_2]$.

1) Limitation (Collinearity): In general, the modulation approach can also be expressed in terms of basis vectors $\mathbf{n}, \mathbf{e}_1, \ldots, \mathbf{e}_d$ and the components of $\Lambda(\mathbf{x})$, λ_i

$$\dot{\mathbf{x}} = \hat{\mathbf{n}}\lambda_1\langle\hat{\mathbf{n}}, \mathbf{v}\rangle + \sum_{i=1}^{d-1} \mathbf{e}_i\lambda_{i+1}\langle\mathbf{e}_i, \mathbf{v}\rangle. \quad (10)$$

Now consider a vector $\mathbf{v} = f_g(\mathbf{x})$, such that $\langle\hat{\mathbf{n}}, \mathbf{v}\rangle = \pm\|\mathbf{v}\|$ [i.e., pointing in the same or opposite direction as the gradient of $\Gamma(\mathbf{x})$]. Then, $\langle\mathbf{e}_1, \mathbf{v}\rangle = \langle\mathbf{e}_2, \mathbf{v}\rangle = 0$, and $\dot{\mathbf{x}} = \pm\hat{\mathbf{n}}\lambda_1\|\mathbf{v}\|$. This causes $\dot{\mathbf{x}} \to 0$, as $\lambda_1 \to 0$, which in turn can cause the robot to not only slow down when the modulation applies, but slowly collide with the obstacle when the dynamics f_g and $\nabla\Gamma(\mathbf{x})$ are coincident. Thus, constructing $M(\mathbf{x})$ based on the current formulation can lead to spurious attractors at the obstacle boundaries, $\partial\mathcal{O}$. This can be addressed in practice by using an external heuristic-based boundary exit strategy [18]; however, this approach is only valid for convex shapes.

2) Improvement (Convergence and Star-Shaped): The presented modulation approach was improved upon by Huber et al. [15] through a closed-form solution that alleviates the stopping problem due to collinearity, while establishing the ability to navigate around nonconvex star-shaped obstacles. This was achieved by introducing a reference point that constructs a nonorthogonal basis $H(\mathbf{x})$. A reference direction, $\mathbf{n}' = \mathbf{x} - \mathbf{r}$, is utilized in lieu of $\mathbf{n} = \nabla\Gamma(\mathbf{x})$ to construct basis $H'(\mathbf{x})$, where $\mathbf{r}$ is a point inside the obstacle region that creates a direction not coincident with any vector in basis $H(\mathbf{x})$.

Interpretation: As $\mathbf{n}'$ is not orthogonal to basis vectors $\mathbf{e}_i$, the method translates $\langle \hat{\mathbf{n}}', \mathbf{v}\rangle$ onto basis axes $\mathbf{e}_i$. The bases can be interchanged by stating $H'(\mathbf{x}) = H(\mathbf{x})\Sigma(\mathbf{x}) \Rightarrow \Sigma(\mathbf{x}) = H(\mathbf{x})^T H'(\mathbf{x})$. $\Sigma(\mathbf{x})$ takes the following form:

$$\Sigma(\mathbf{x}) = \begin{bmatrix} \hat{\mathbf{n}}^T\hat{\mathbf{n}}' & 0 & \cdots & 0 \\ \mathbf{e}_1^T\hat{\mathbf{n}}' & 1 & & 0 \\ \vdots & & \ddots & \\ \mathbf{e}_{d-1}^T\hat{\mathbf{n}}' & 0 & \cdots & 1 \end{bmatrix}.$$

The resulting modulation, $M'(\mathbf{x}) = H'(\mathbf{x})\Lambda(\mathbf{x})H'(\mathbf{x})^{-1}$, can be related to orthonormal basis $H(\mathbf{x})$ through $M'(\mathbf{x}) = H(\mathbf{x})\Lambda'(\mathbf{x})H(\mathbf{x})^T, \Lambda' = \Sigma\Lambda\Sigma^{-1}$. The effective scaling matrix $\Lambda'(\mathbf{x})$, can, therefore, be shown as the following:

$$\Lambda'(\mathbf{x}) = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ \frac{\mathbf{e}_1^T\hat{\mathbf{n}}'(\lambda_1-\lambda_2)}{\hat{\mathbf{n}}^T\hat{\mathbf{n}}'} & \lambda_2 & & 0 \\ \vdots & & \ddots & \\ \frac{\mathbf{e}_{d-1}^T\hat{\mathbf{n}}'(\lambda_1-\lambda_d)}{\hat{\mathbf{n}}^T\hat{\mathbf{n}}'} & 0 & \cdots & \lambda_d \end{bmatrix}. \quad (11)$$

As such, the reference point method [15] creates off-diagonal components in the $\Lambda(\mathbf{x})$ matrix. The velocity of the modulated system can be stated as follows:

$$\dot{\mathbf{x}} = M(\mathbf{x})f_g(\mathbf{x}) - \frac{2}{\Gamma(\mathbf{x})}\sum_{i=1}^{d-1}\mathbf{e}_i\frac{\mathbf{e}_i^T\hat{\mathbf{n}}'}{\hat{\mathbf{n}}^T\hat{\mathbf{n}}'}\hat{\mathbf{n}}^T f_g(\mathbf{x}). \quad (12)$$

This method has been proven to preserve stability and convergence via contraction theory almost everywhere except at a single saddle point trajectory (i.e., when the reference direction is collinear to original velocity) leads to a spurious attractor. Note that as $\langle \hat{\mathbf{n}}, \hat{\mathbf{n}}'\rangle \to 0, \dot{\mathbf{x}} \to \infty$, requiring that care be taken with reference point placement in the context of nonconvex obstacles, such as those introduced in this work. In general, it can be difficult to place reference points, such that the resulting velocity guarantees circumnavigation of a nonconvex obstacle.

3) Continuity: As illustrated in Fig. 3, modulation strategies rely on the construction of a tangent plane $\mathbf{E} = [\mathbf{e}_1 \;\; \mathbf{e}_2 \;\; \ldots \;\; \mathbf{e}_{d-1}]$ that is orthogonal to the obstacle boundary $\partial\mathcal{O}$ defined by $\mathbf{n} = \nabla\Gamma(\mathbf{x})$. Note that $\Gamma(\mathbf{x}) = 1$ (the obstacle boundary) and its level-sets $\Gamma(\mathbf{x}) = \gamma$ with $\gamma \in \mathbb{R}_{\geq 1}$ are smooth manifolds, $\mathcal{M}$, which we define as γ-manifolds.

Definition 1 (γ-manifold): Let a $\mathcal{C}^2$ differentiable scalar-valued field $\Gamma(\mathbf{x}) : \mathbb{R}^d \to \mathbb{R}$ describe the proximity of the robot state to the obstacle boundary $\partial\mathcal{O}$. The γ-manifold is the $(d-1)$-dimensional manifold, $\mathcal{M}_\gamma$ representing the isosurface of $\Gamma(\mathbf{x}) = \gamma$ with $\gamma \in \mathbb{R}_{\geq 1}$.

Then, every tangent vector $\mathbf{e}_i(\mathbf{x})$ defined on the space of a γ-manifold is in the image of a tangent vector field $\mathbf{e}_i(\mathbf{x})$ on $\mathcal{M}_\gamma$. The DS modulation approach assumes obstacles that are bounded, convex, and star-shaped; as such, for any continuous vector field on M_γ in $\mathbb{R}^3$, its image must contain at least one tangent vector that vanishes. This is due to the *hairy ball theorem* [5] from algebraic topology.

Theorem 1 (Hairy Ball Theorem): Let $\mathbb{S}^n = \{\mathbf{p} \in \mathbb{R}^{n+1} : |\mathbf{p}| = 1\}$. If $\mathbf{v}$ is a continuous vector field on $\mathbb{S}^n$, with $n \in \{2, 4, 6, \ldots\}$, then, there exists $\mathbf{p} \in \mathbb{S}^n$ such that $\mathbf{v}(\mathbf{p}) = \mathbf{0}$.

Proof: See McGrath [25]. ■

This implies that for all 3-D obstacles that are spheres $\mathbb{S}^2$ or diffeomorphic to $\mathbb{S}^2$ (which include all connected, compact, convex, and star-shaped obstacles), there must be at least one point on $\mathcal{M}_\gamma$ where the tangent vector $\mathbf{e}_i(\mathbf{x})$ vanishes. Hence, the tangent vector field on $\mathcal{M}_\gamma$ must be discontinuous. While this suggests that the resulting motion policy is discontinuous as well, we demonstrate that this is not the case below.

Let $T(\mathbf{n})$ be a tangent plane projection operator, and let $\mathbf{E}$ be the set of orthogonal vectors describing the tangent plane. Then, $T(\mathbf{n})$ can be expressed as

$$T(\mathbf{n}) = \mathbb{I} - \frac{\mathbf{n}\mathbf{n}^T}{\|\mathbf{n}\|^2} = \mathbf{E}\mathbf{E}^T. \quad (13)$$

This follows from the outerproduct representation of the basis $H(\mathbf{x})H(\mathbf{x})^T$, wherein we note that:

$$H(\mathbf{x})H(\mathbf{x})^T = \hat{\mathbf{n}}\hat{\mathbf{n}}^T + \mathbf{E}\mathbf{E}^T = \mathbb{I}.$$

The modulation approach of Zadeh and Billard [18] can, thus, be expressed as a function of $\Gamma(\mathbf{x})$ and $\mathbf{n} = \nabla\Gamma(\mathbf{x})$

$$\dot{\mathbf{x}} = \left(\left(1 - \frac{1}{\Gamma(\mathbf{x})}\right)\frac{\mathbf{n}\mathbf{n}^T}{\|\mathbf{n}\|^2} + \left(1 + \frac{1}{\Gamma(\mathbf{x})}\right)T(\mathbf{n})\right)f_g(\mathbf{x})$$
$$= \left(\mathbb{I} + \frac{\mathbb{I} - 2\cdot\hat{\mathbf{n}}\hat{\mathbf{n}}^T}{\Gamma(\mathbf{x})}\right)f_g(\mathbf{x}). \quad (14)$$

Note that when $\Gamma(\mathbf{x}) = 1$, then, (14) can be expressed as

$$\dot{\mathbf{x}} = 2\cdot T(\mathbf{n})f_g(\mathbf{x}). \quad (15)$$

This demonstrates that the modulation acts as a tangent plane projection when on the obstacle boundary. As $f_g(\mathbf{x})$, $\Gamma(\mathbf{x})$, and $\nabla\Gamma(\mathbf{x})$ are continuous, so are the modulation strategies (10) and (12). Nevertheless, as shown previously, spurious equilibrium points may arise when the nominal dynamics $f_g(\mathbf{x})$ and the normal $\hat{\mathbf{n}}$ (or reference $\hat{\mathbf{n}}'$) directions are collinear. Such spurious equilibrium points arise in $\mathbb{R}^{n+1}$ with $n \in \{2, 4, 6, \ldots\}$, due to the *hairy ball theorem* [5], [25].

V. On-Manifold DS Modulation

State-of-the-art modulation strategies are only capable of circumnavigating star-shaped nonconvex obstacles. In addition, the reference point technique introduced by Huber et al. [15] requires explicit placement of reference points, which can be difficult to do in a manner that ensures obstacle circumnavigation. In this section, we introduce an approach to constructing modulation matrix $M(\mathbf{x})$ that:

1) avoids the generation of spurious attractors arising from collinearity (i.e., stopping on the edge of an obstacle when $\mathbf{n}$ and $f_g(\mathbf{x})$ are collinear);
2) enables isosurface tracking; and
3) enables introduction of on-manifold strategies for circumnavigating nonconvex obstacles.

The methodology discussed here modifies modulation $M(\mathbf{x})$ to be a function of $f_g(\mathbf{x})$.

A. Partial On-Manifold Dynamics and Spurious Attractors

In this work, we consider the isosurfaces of $\Gamma(\mathbf{x})$ with different values of $\gamma > 1 \in \mathbb{R}$, i.e., the γ-manifolds introduced in Definition 1, as $(d-1)$-manifolds, which can be circumnavigated or used as level-sets to navigate between manifolds (e.g., from $\gamma = 1$ to $\gamma > 1$ to escape nonconvex regions as will be described in Section V-B. Due to the $\mathcal{C}^2$-differentiability requirement we enforced on the obstacle representation function, $\Gamma(\mathbf{x})$, the γ-manifolds will always be smooth; i.e., no discontinuities. From herein, we drop the γ- and refer to such γ-manifolds simply as manifolds.

From (11), we can construct an explicitly decomposed modulation for the off-diagonal elements as follows:

$$\Lambda'(\mathbf{x}) = \Lambda(\mathbf{x}) + \begin{bmatrix} 0 \\ \mathbf{E}^T\phi(\mathbf{x}) \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0} \end{bmatrix} \tag{16}$$

with $\phi(\mathbf{x}) \in \mathbb{R}^{d\times 1}$ and $\mathbf{0} = \mathbf{0}_{1\times(d-1)}$. Here, $\phi(\mathbf{x})$ is an on-manifold dynamics term: an auxilary direction in Euclidean space that is projected onto the tangent plane [i.e., $\phi'(\mathbf{x}) = \mathbf{E}(\mathbf{x})^T\phi(\mathbf{x})$]. Within these equations, $\mathbf{E} = [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \dots \quad \mathbf{e}_{d-1}] \in \mathbb{R}^{d\times(d-1)}$ is the collection of basis vectors orthogonal to $\nabla\Gamma(\mathbf{x})$ that are used to construct orthonormal basis $H(\mathbf{x})$.

By inserting (16) into the classical modulation approach [18] defined in (7), with $H(\mathbf{x})$ defined as in (8), we obtain the following modulation matrix formulation:

$$\begin{aligned} M'(x) &= H(\mathbf{x})\Lambda'(\mathbf{x})H(\mathbf{x})^T \\ &= M(\mathbf{x}) + H(\mathbf{x}) \begin{bmatrix} 0 \\ \mathbf{E}^T\phi(\mathbf{x}) \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0} \end{bmatrix} H(\mathbf{x})^T \end{aligned} \tag{17}$$

Incorporating $H(\mathbf{x})$ from (8) in the second term in (17) yields $H(\mathbf{x})[\begin{smallmatrix} 0 \\ \mathbf{E}^T\phi(\mathbf{x}) \end{smallmatrix}][1 \quad \mathbf{0}]H(\mathbf{x})^T = \mathbf{E}\mathbf{E}^T \cdot \phi(\mathbf{x})\mathbf{n}^T$. To understand this approach, consider the isosurface of $\Gamma(\mathbf{x}) = \gamma$ to be a *smooth manifold*, $\mathcal{M}$, with an associated tangent space $T_p\mathcal{M}$, described by $\mathbf{E}$. The effect of introducing $\phi(\mathbf{x})$ is to induce on-manifold dynamics, scaled by $\langle \mathbf{n}, f_g(\mathbf{x})\rangle$, using the manifold-projection matrix $\mathbf{E}\mathbf{E}^T$

$$\begin{aligned} \dot{\mathbf{x}} &= M'(\mathbf{x})f_g(\mathbf{x}) \\ &= M(\mathbf{x})f_g(\mathbf{x}) + \mathbf{E}\mathbf{E}^T\phi(\mathbf{x})\mathbf{n}^T f_g(\mathbf{x}). \end{aligned} \tag{18}$$

Hence, if $\phi(\mathbf{x}) \neq \mathbf{0}$ any spurious attractors stemming from collinearity of $\langle \hat{\mathbf{n}}, \mathbf{v}\rangle = \pm\|\mathbf{v}\|$ with $\mathbf{v} = f_g(\mathbf{x})$ are never introduced, as the second term in (18) will induce motion in the tangent space. Thus, to remove spurious attractors on the boundary, we need only set $\phi(\mathbf{x}) = \delta \cdot \psi(\langle \mathbf{n}, \mathbf{v}\rangle) \cdot \mathbf{e}_i, i \in \{1, \dots, (d-1)\}$, where $\psi(\zeta)$ acts to enable the effect when $\zeta \to -1$, and disable it otherwise. For this purpose, we propose $\psi_k(\zeta) = \max(0, (-\zeta)^k)$, with $k \in [1, 3, 5, \dots]$. To ensure that the effect is only active at low values of $\Gamma(\mathbf{x})$, we construct $\phi(\mathbf{x})$ such that $\Gamma(\mathbf{x}) \to \infty \Rightarrow M(\mathbf{x}) = \mathbb{I}$

$$\phi(\mathbf{x}) = \begin{cases} \delta/\Gamma(\mathbf{x})^2 \cdot \psi_k(\langle \mathbf{n}, \mathbf{v}\rangle) \cdot \mathbf{e}_i, & \langle \mathbf{e}_i, \mathbf{v}\rangle \geq 0 \\ \mathbf{0}, & \text{otherwise.} \end{cases} \tag{19}$$

This introduces our modulation strategy as a piecewise smooth (PWS) dynamical system [9] governed by the switching function $\langle \mathbf{e}_i, \mathbf{v}\rangle$, with the first term in (18) being continuous everywhere (as demonstrated in Section IV-C3) and the second term subject to the discontinuity shown in (19). Interestingly, it resembles the form of a Filippov system [12] with a discontinuous set defined by $\langle \mathbf{e}_i, \mathbf{v}\rangle \geq 0$. Note that (18) can be rewritten as a decomposition of velocity along $\nabla\Gamma(\mathbf{x})$ and the effective tangent space velocity as

$$\begin{aligned} \dot{\mathbf{x}} &= M'(\mathbf{x})f_g(\mathbf{x}) = H(\mathbf{x}) \begin{bmatrix} \dot{\mathbf{x}}_{\mathbf{n}} & \dot{\mathbf{x}}_{\mathbf{E}} \end{bmatrix}^T \\ &= H(\mathbf{x}) \begin{bmatrix} \Lambda_{\mathbf{n}}\mathbf{n}^T f_g(\mathbf{x}) \\ \mathbf{E}^T\phi(\mathbf{x})\mathbf{n}^T f_g(\mathbf{x}) + \Lambda_{\mathbf{E}}\mathbf{E}^T f_g(\mathbf{x}) \end{bmatrix}. \end{aligned} \tag{20}$$

This follows from $\Lambda' = [\begin{smallmatrix} \Lambda_{\mathbf{n}} & 0 \\ \phi(\mathbf{x}) & \Lambda_{\mathbf{E}} \end{smallmatrix}]$. In order to induce tangent space velocities at the collinearity cases (and consequently avoid spurious attractors) one must ensure that $(\Lambda_{\mathbf{E}}\mathbf{E}^T + \mathbf{E}^T\phi(\mathbf{x})\mathbf{n}^T) \neq \mathbf{0}_{d-1\times d}$. The effect of this method on the spurious attractor arising due to collinearity is illustrated in Fig. 4 for a spherical obstacle and linear DS $f_g(\mathbf{x})$. *Interpretation:* Following the hairy ball theorem, and our analysis in Section IV-C3, a nonzero continuous tangent vector field does not exist for obstacle boundaries diffeomorphic to even-dimensional spheres ($\mathbb{S}^{2,4,\cdots}$), implying one or more spurious attractors depending on how the discontinuous or vanishing vector field is designed. Hence, to remove a spurious attractor we artificially introduce a single *discontinuity* via $\mathbf{e}_i \in \phi(\mathbf{x})$ on the obstacle boundary when $\langle \hat{\mathbf{n}}, \hat{\mathbf{v}}\rangle = 1$ and, therefore, $\|\phi(\mathbf{x})\| = \psi_k(1) = 0$. Even though our novel on-manifold modulation strategy results in a PWS DS impenetrability is preserved and, as opposed to prior methods, global convergence is achieved.

Theorem 2 (Impenetrability): Consider an obstacle with arbitrary shape $\mathcal{O} \in \mathbb{R}^d$ represented by a function $\Gamma(\mathbf{x}) : \mathbb{R}^d \to \mathbb{R}$ as defined in (3)–(5). Any trajectory $\mathbf{x}(t), t \in [0, \infty)$, that starts outside the obstacle boundary, i.e., $\Gamma(\mathbf{x}(0)) \geq 1$, and evolves according to dynamics with modulation constructed by (18) and arbitrary DS $f_g(\mathbf{x})$, will never penetrate the obstacle; i.e., $\Gamma(\mathbf{x}(t)) \geq 1, t \in (0, \infty)$.

Proof: See Appendix A. ■

With the presented approach impenetrability is guaranteed for arbitrary nonconvex obstacles and arbitrary DS $f_g(\mathbf{x})$; i.e., $f_g(\mathbf{x})$ can be linear, nonlinear, multitarget, limit cycle, etc. As will be shown, convergence toward a unique equilibrium point $\mathbf{x}^* \in \mathbb{R}^N$ for nonconvex obstacles relies on convergence for convex obstacles. Further, as in [15], we assume the global DS to be modulated, $f_g(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}^*)$, is a linear system that is globally asymptotically stable toward a single attractor $\mathbf{x}^* \in \mathbb{R}^N$ ensured via Lyapunov theory.

Theorem 3 (Convergence for Convex Obstacles): Consider a linear DS $f_g(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}^*)$ and a convex obstacle $\mathcal{O} \in \mathbb{R}^d$

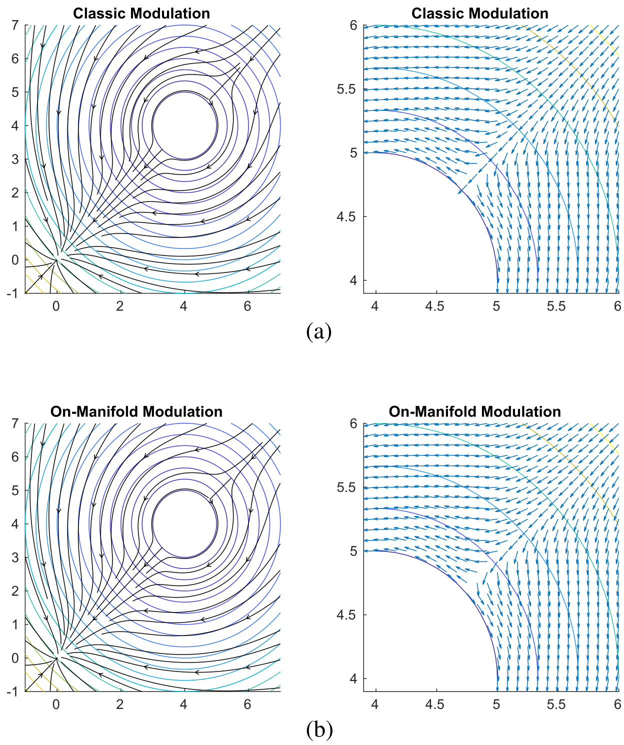


Fig. 4. Spurious attractor due to collinearity of $\mathbf{n}$ and $f_g(\mathbf{x})$ can be eliminated using off-diagonal elements in the $\Lambda(\mathbf{x})$ matrix. Classic modulation is shown in Fig. 4(a) and on-manifold modulation in Fig. 4(b). Note how overall modulation performance remains unaffected (left figures), while the spurious attractor is removed (right figures). (a) Classic modulation (as demonstrated in [18]). Collinearity of $\mathbf{n}$ and $f_g(\mathbf{x})$ leads to a spurious attractor as illustrated on the right. (b) On-manifold modulation [applying $\phi(\mathbf{x})$ following (19)] leads to velocities in the tangent space of $\mathcal{M}$, eliminating the spurious attractor associated with collinearity, $\delta = 0.1, k = 1001$.

represented by $\Gamma(\mathbf{x}) : \mathbb{R}^d \to \mathbb{R}$ as in (3)–(5). If $f_g(\mathbf{x})$ is modulated according to (18) around the boundary $\Gamma(\mathbf{x}) = 1$, then, any trajectory $\mathbf{x}(t), t \in [0, \infty]$, that starts outside the obstacle boundary, i.e., $\Gamma(\mathbf{x}(0)) \geq 1$, converges to $\mathbf{x}^*$ if there exists $\mathbf{P} \succ 0$ such that $\mathbf{n}^T\mathbf{P}\mathbf{e}_i > 0$, with $\succ 0$ denoting positive definiteness of a matrix.

Proof: See Appendix B. ∎

As shown in Theorems 2 and 3, by introducing tangent space dynamics into the modulation strategy we can ensure impenetrability and convergence for convex obstacles without inducing any saddle points, local minima, or spurious attractors, and without the need for a reference direction (Contribution 1). Furthermore, similar to Alvarez et al. [1] in the proof of Theorem 3 we show that the discontinuity introduced by $\phi(\mathbf{x})$ in (19) does not hinder global stability and convergence to $\mathbf{x}^*$, in fact, it is necessary due to the hairy ball theorem. To preserve the guarantees for arbitrary nonconvex obstacles, we present a novel approach in Section V-B that tracks isosurfaces $\Gamma(\mathbf{x}) = \gamma > 1$ that can be approximated as convex shapes and used to circumnavigate nonconvex regions. Further, in Section V-C, we present a suite of local on-manifold navigation strategies for highly concave regions. These approaches follow and extend the modulation strategy presented in (16)–(18), yet are applicable to nonconvex shapes by redefining and reparameterizing $\Lambda(\mathbf{x})$ and $\phi(\mathbf{x})$.

B. Isosurface Tracking

Consider a form of $M(\mathbf{x}) = H(\mathbf{x})\Lambda'(\mathbf{x})H(\mathbf{x})^T$, such that

$$\Lambda'(\mathbf{x}) = \Lambda(\mathbf{x}) + \begin{bmatrix}\lambda'_{11} & \mathbf{0}\end{bmatrix}^T \begin{bmatrix}1 & \mathbf{0}\end{bmatrix} \tag{21}$$

$\Lambda(\mathbf{x})$ is identical to that introduced in (9). Consider the velocity along the gradient of $\mathbf{n}$, $\dot{\mathbf{x}}_{\mathbf{n}}$ (such that $\dot{\mathbf{x}} = H(\mathbf{x})[\dot{\mathbf{x}}_{\mathbf{n}} \quad \dot{\mathbf{x}}_{\mathbf{E}}]^T$), for this new system

$$\dot{\mathbf{x}}_{\mathbf{n}} = c\left(1 - \frac{1}{\Gamma(\mathbf{x})} + \lambda'_{11}\right)\mathbf{n}^T f_g(\mathbf{x}) \tag{22}$$

c is a term introduced to maintain convergence properties. Solving for λ'_{11} at $\dot{\mathbf{x}}_{\mathbf{n}} = 0$, we can cause the system to track isosurface $\Gamma(\mathbf{x}) = \gamma$ by setting the following:

$$\lambda'_{11} = \frac{1}{\gamma} - 1. \tag{23}$$

The constant c can be set to γ to maintain the original system's convergence properties (i.e., that $\Gamma(\mathbf{x}) \to \infty \Rightarrow M(\mathbf{x}) \to \mathbb{I}$); however, the sign of $\mathbf{n}^T f_g(\mathbf{x})$ affects the system's stability. $\mathbf{n}^T f_g(\mathbf{x}) < 0$ will result in exact manifold tracking, but $\mathbf{n}^T f_g(\mathbf{x}) > 0$ represents an unstable dynamical system, which will cause movement toward the obstacle boundary when $\Gamma(\mathbf{x}) < \gamma$. To correct for this, we introduce function $\varphi(\Gamma(\mathbf{x}), \gamma, \mathbf{u}, f_g(\mathbf{x}))$, referred to as $\varphi(\mathbf{u})$ to ease notation

$$\varphi(\mathbf{u}) = \begin{cases} -1, & \Gamma(\mathbf{x}) < \gamma \wedge \mathbf{u}^T \hat{f}_g(\mathbf{x}) > 0 \\ 1, & \text{otherwise.} \end{cases} \tag{24}$$

The function can be leveraged in (21) as follows:

$$\lambda'_{11} = \gamma \cdot \varphi(\mathbf{n})\left(\frac{1}{\gamma} - \frac{1}{\Gamma(\mathbf{x})}\right) - \left(1 - \frac{1}{\Gamma(\mathbf{x})}\right). \tag{25}$$

Fig. 5(a) depicts the effect of this, and demonstrates how this methodology can enable partial isosurface tracking. Note that in this format, partial isosurface tracking ensures velocity $\dot{\mathbf{x}}_n \geq 0$ when $\Gamma(\mathbf{x}) \leq \gamma$. Isosurface tracking effectively shifts the boundary of the obstacle to $\Gamma(\mathbf{x}) = \gamma$, maintaining the impenetrability and convergence guarantees of the original system.

Proposition 1 (Impenetrability and Convergence of Shifted Boundary): Any trajectory $\{\mathbf{x}\}_t$, that starts outside the convex shifted obstacle boundary, i.e., $\Gamma(\mathbf{x}) \geq \gamma > 1$, and evolves according to dynamics with modulation matrix constructed following (21) and (25) and assuming a linear DS with a single attractor at $\mathbf{x}^* \in \mathbb{R}^d$, will never penetrate the shifted boundary; i.e., $\Gamma(\{\mathbf{x}\}_t) \geq \gamma, t = 0 \ldots \infty$, and is ensured to converge to the attractor of the linear DS $\mathbf{x}^*$.

Proof: Follows from Theorems 2 and 3. ∎

Care should be taken when selecting tracking level $\Gamma(\mathbf{x}) = \gamma$; conservative selection can lead to disconnected regions, constraining the robot to a subspace of the problem area, whereas low values can lead to a nonconvex manifold at $\Gamma(\mathbf{x}) = \gamma$, requiring complex reasoning to ensure circumnavigation of the obstacle (we discuss such techniques in Section V-C). Note that

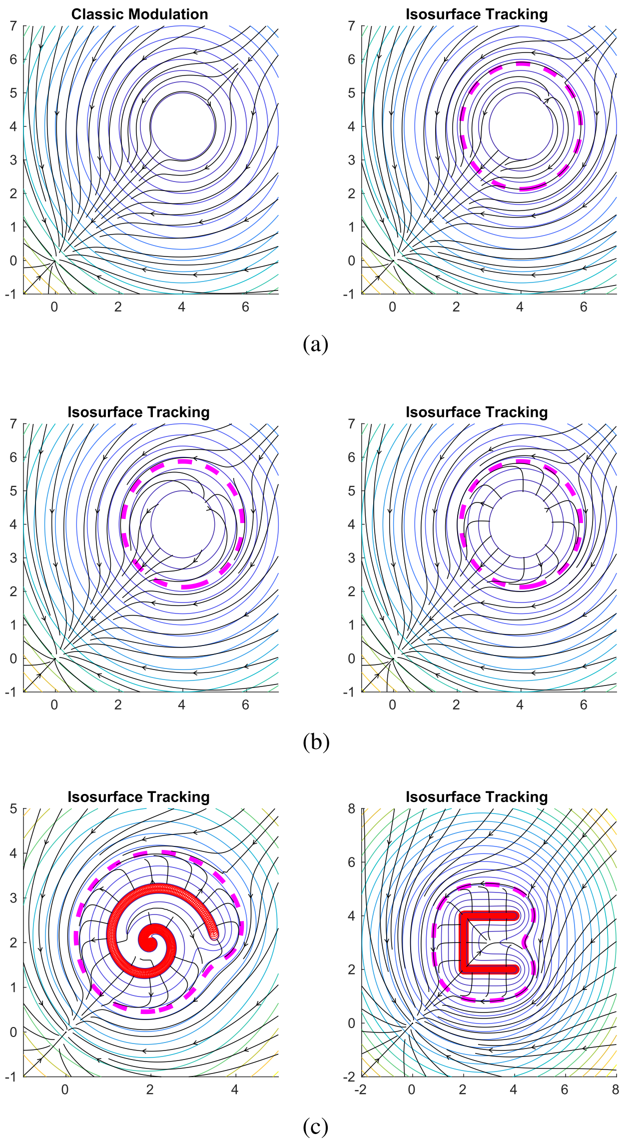


Fig. 5. Elements of isosurface tracking. By combining tangent space navigation with gradient ascent, specific isosurfaces can be tracked upon which local exit strategies are employed for circumnavigation. For details about the obstacle exit strategies see Section V-C. (a) Partial isosurface tracking: an offset on λ_{11} enables isosurface tracking, although a correction must be employed to ensure stability. (Left) Modulation without isosurface tracking; (Right) Modulation with partial isosurface tracking (the selected surface is highlighted in purple). (b) Full isosurface tracking: c_γ affects the width of the tracking band, and can be used to balance gradient ascent with induced tangent space dynamics. (Left) $c_\gamma = 10$ (Right) $c_\gamma = 1000$. (c) Combination of isosurface tracking with a narrow band (i.e., gradient ascent) and the obstacle exit strategy methodology allows for nonconvex obstacles to be represented and circumnavigated using a local strategy.

the selected tracking level should be lower than that of the target attractor (i.e., that $\Gamma(\mathbf{x}_f) \leq \gamma$).

Nonconvex Obstacles: While the isosurface tracking approach works well for tracking convex isosurfaces from outside the shifted boundary (i.e., when $\Gamma(x) \geq \gamma > 1$), for nonconvex obstacle boundaries we must also modify the system dynamics in the tangent space within the internal boundary region between $\Gamma(x) = \gamma$ and $\Gamma(x) = 1$. The strategy here is to override the dynamics internal to the shifted boundary in order to follow the path of steepest ascent of $\Gamma(\mathbf{x})$ (i.e., when $\Gamma(\mathbf{x}) < \gamma$). When on the shifted boundary ($\Gamma(\mathbf{x}) = \gamma$), we wish for the tangent space dynamics induced by $\phi(\mathbf{x})$ to navigate around the obstacle; when far from the obstacle ($\Gamma(\mathbf{x}) > \gamma$) we seek behavior similar to the classic modulation strategy [18]. Thus, we introduce three different $\Lambda(\mathbf{x})$ matrices to accomplish the desired behavior

$$\Lambda(\mathbf{x}) = \Lambda_\nabla(\mathbf{x}) + \Lambda_\phi(\mathbf{x}) + \varphi_\Gamma(\mathbf{x})\Lambda_\Gamma(\mathbf{x}). \tag{26}$$

Here, $\Lambda_\nabla(\mathbf{x})$ encodes the gradient ascent dynamics, $\Lambda_\phi(\mathbf{x})$ the tangent space dynamics, and $\Lambda_\Gamma(\mathbf{x})$ the classic modulation behavior [the behavior arising from (7)–(9)]. Within the boundary, navigating to the target manifold can be accomplished through a modulation matrix of the following form:

$$\Lambda_\nabla(\mathbf{x}) = \begin{bmatrix} 1 \\ \mathbf{0} \end{bmatrix} \begin{bmatrix} \varphi(\mathbf{n}) & \varphi'(\mathbf{e}_1) & \dots & \varphi'(\mathbf{e}_{d-1}) \end{bmatrix} \lambda_{11} \tag{27}$$

$\lambda_{11} = \gamma(1/\gamma - 1/\Gamma(\mathbf{x}))$ and $\varphi'(\mathbf{u})$ is a similar function to $\varphi(\mathbf{u})$ but is only active internal to the shifted boundary

$$\varphi'(\mathbf{u}) = \begin{cases} -1, & \Gamma(\mathbf{x}) < \gamma \wedge \mathbf{u}^T \hat{f}_g(\mathbf{x}) > 0 \\ 1, & \Gamma(\mathbf{x}) < \gamma \wedge \mathbf{u}^T \hat{f}_g(\mathbf{x}) < 0 \\ 0, & \text{otherwise} \end{cases} \tag{28}$$

$\Lambda_\nabla(\mathbf{x})$ alone will cause the system to track the isosurface at $\Gamma(\mathbf{x}) = \gamma$ using both normal and tangential velocity components.

Theorem 4 (Convergence to Isosurface for Gradient Ascent): Let $\Gamma(\mathbf{x})$ be defined according to (3)–(5), with C^2 differentiability and $\|\nabla\Gamma(\mathbf{x})\| > 0 \forall \mathbf{x} \notin \mathcal{O}$. A path must then exist from every point on $\Gamma(\mathbf{x}) = 1$ to a point on $\Gamma(\mathbf{x}) = \gamma$ for which $\Gamma(\mathbf{x})$ is monotonically increasing, and $\|\nabla\Gamma(\mathbf{x})\| > 0$.

Proof: As $\nabla\Gamma(\mathbf{x})$ is smooth [(3)–(5)], this implies that no saddle points or local maxima exist, and that following $\nabla\Gamma(\mathbf{x})$ will lead to the selected isosurface for any point $\mathbf{x} \in \{\mathbf{x} : \Gamma(\mathbf{x}) < \gamma\} \cap \{\mathbf{x} : \mathbf{x} \notin \mathcal{O}\}$. ■

When $\Gamma(\mathbf{x}) > \gamma$, only the velocity component in the direction of $\nabla\Gamma(\mathbf{x})$ is scaled (similar to that in the classic modulation). Note that the induced normal velocity is zero on the boundary. To ensure that dynamics emanating from the obstacle are not suppressed, we augment λ_{11} with a small value ϵ when $\mathbf{n}$ and $\mathbf{v}$ are collinear

$$\lambda_{11} = \gamma\left(\frac{1}{\gamma} - \frac{1}{\Gamma(\mathbf{x})}\right) + \frac{\epsilon \cdot \psi(\langle \mathbf{n}, \mathbf{v} \rangle)}{\Gamma(\mathbf{x})^2}. \tag{29}$$

Note that ϵ is scaled by $1/\Gamma(\mathbf{x})^2$; this is a design choice to maintain the properties of the modulation (i.e., that $\Gamma(\mathbf{x}) \gg 1 \Rightarrow M(\mathbf{x}) \to \mathbb{I}$).

However, while $\Lambda_\nabla(\mathbf{x})$ will ensure convergence to the secondary obstacle boundary, it negates motion in the tangent space of $\Gamma(\mathbf{x})$ (used alone, the system will come to rest on the boundary of $\Gamma(\mathbf{x}) = \gamma$). Here, we elect to utilize on-manifold dynamics encoded by $\phi(\mathbf{x})$. The motivation for $\phi(\mathbf{x})$ to be the primary source of system dynamics is based on the merits of tangent-space navigation. Of particular significance is that in dense environments involving nonconvex obstacles, the isosurface selected may be lower than one that is explicitly convex, allowing the use of $\phi(\mathbf{x})$

as a means to locally navigate around nonconvex obstacles while remaining tangential to the isosurface. We define the tangent space navigation matrix as

$$\Lambda_\phi(\mathbf{x}) = \frac{1}{1 + c_\gamma(\Gamma(\mathbf{x}) - \gamma)^2} \cdot \operatorname{diag}\left(\mathbf{E} \cdot \phi(\mathbf{x})\right) \frac{\mathbf{1}\mathbf{v}^T H}{\|\mathbf{v}\|}. \quad (30)$$

Note that this replaces the dynamics $\mathbf{v} = f_g(\mathbf{x})$ with the tangent space dynamics described by $\phi(\mathbf{x})$ when $\Gamma(\mathbf{x}) = \gamma$, as $\mathbf{v}^T H H^T \mathbf{v} = \|\mathbf{v}\|^2$ (as $HH^T = H^T H = \mathbb{I}$). c_γ is a constant that can be used to control the width of the tracking band (i.e., where dynamics induced by $\phi(\mathbf{x})$ are active). This constant should be chosen based on the desired behavior: larger values will only enable on-manifold dynamics, $\phi(\mathbf{x})$, when close to the tracking level, $\Gamma(\mathbf{x}) = \gamma$.

Finally, to ensure classic modulation further from the obstacle, we define the regular dynamics matrix as follows:

$$\Lambda_\Gamma(\mathbf{x}) = \operatorname{diag}\left(\begin{bmatrix} 0 & \mathbf{1} \end{bmatrix}\right) C_\Gamma(\mathbf{x}) \left(1 + \frac{1}{\Gamma(\mathbf{x})}\right) \quad (31)$$

The additional term $C_\Gamma(\mathbf{x})$ acts to smoothly blend the traditional modulation with $\phi(\mathbf{x})$ as $\Gamma(\mathbf{x})$ approaches the target isosurface

$$C_\Gamma(\mathbf{x}) = \frac{c_\gamma(\Gamma(\mathbf{x}) - \gamma)^2}{1 + c_\gamma(\Gamma(\mathbf{x}) - \gamma)^2}. \quad (32)$$

Depending upon the desired behavior, $\Lambda_\Gamma(\mathbf{x})$ can be disabled within the boundary using $\varphi_\Gamma(\mathbf{x}) = \Gamma(\mathbf{x}) > \gamma$. Combining each of these matrices results in the final modulation as follows:

$$\Lambda(\mathbf{x}) = \Lambda_\nabla(\mathbf{x}) + \Lambda_\phi(\mathbf{x}) + \varphi_\Gamma(\mathbf{x})\Lambda_\Gamma(\mathbf{x}).$$

This is utilized in standard form: $M(\mathbf{x}) = H(\mathbf{x})\Lambda(\mathbf{x})H(\mathbf{x})^T$. Fig. 5(b) demonstrates how the constant c_γ can be adapted to tradeoff gradient ascent and the obstacle exit strategy.

Theorem 5 [Convergence to Attractor for (26)]: Let $\phi(\mathbf{x})$ represent a consistent, on-manifold direction to a point $\mathbf{x}'$, such that $\hat{\mathbf{n}} \cdot \hat{\mathbf{v}} = 1, \mathbf{n} = \nabla\Gamma(\mathbf{x})$, and $\mathbf{v} = f_g(\mathbf{x})$. Assume $\mathbf{x}_f$ is the settling point for the input dynamics (i.e., that $\mathbf{x}_f = \lim_{t\to\infty} \int_0^t f_g(\mathbf{x})dt + \mathbf{x}_0$), that input dynamics $f_g(\mathbf{x})$ are linear, and that $\Gamma(\mathbf{x}_f) > \gamma$. Then, (26) preserves the convergence properties of the original system $f_g(\mathbf{x})$.

Proof: See Appendix C. ∎

Fig. 5(c) demonstrates how the methodology can be applied to nonconvex obstacles, where the selection of an isosurface and use of gradient ascent enables escape from nonconvex obstacles, whereas the obstacle exit strategy can be employed to circumnavigate a nonconvex isosurface.

C. Obstacle Exit Strategies

On-manifold dynamics, through the introduction of off-diagonal elements in the $\Lambda(\mathbf{x})$ matrix, can enable sophisticated local navigation capabilities. In this section, we highlight obstacle exit strategies that can be employed to efficiently circumnavigate both convex and nonconvex obstacles. Note that $\phi(\mathbf{x})$ operates within the tangent space, requiring any obstacle exit strategy operating in Euclidean space to be transformed to the tangent space. We define $\phi(\mathbf{x})$ as the obstacle exit strategy in Euclidean space that can be transformed to the tangent space by $\phi'(\mathbf{x}) = E^T\phi(\mathbf{x})$; this requires all $\phi(\mathbf{x})$ methodologies to produce vectors that are not in the nullspace of $\mathbf{E}$—i.e., that $\phi(\mathbf{x}) \notin \mathcal{N}(\mathbf{E}) \Rightarrow |\phi(\mathbf{x}) \cdot \nabla\Gamma(\mathbf{x})|/(\|\phi(\mathbf{x})\|\|\nabla\Gamma(\mathbf{x})\|) < 1$. Next, we introduce three obstacle exit strategies for use in on-manifold circumnavigation. Note that all methodologies assume the attractor position is outside of the shifted boundary (i.e., that $\Gamma(\mathbf{x}_f) > \gamma, \mathbf{x}_f = \lim_{t\to\infty} \int_0^t f_g(\mathbf{x})dt + \mathbf{x}_0$).

1) Hessian: The Hessian of $\Gamma(\mathbf{x})$, notated as $\nabla^2\Gamma(\mathbf{x})$ in this text, enables a gradient ascent methodology for circumnavigating nonplanar convex isosurfaces. Consider a point of divergence from the obstacle surface; at this point $\mathbf{x}_t$, the dynamics must satisfy $\hat{\mathbf{n}} \cdot \hat{\mathbf{v}} > 0, \mathbf{n} = \nabla\Gamma(\mathbf{x}_t), \mathbf{v} = f_g(\mathbf{x}_t)$. We can consider an optimal exit point on the obstacle to be of the following form:

$$C(\mathbf{x}) = \nabla\Gamma(\mathbf{x}) \cdot f_g(\mathbf{x}). \quad (33)$$

In order to maximize (33), the gradient can be used as the on-manifold direction, $\phi(\mathbf{x})$

$$\phi(\mathbf{x}) = \nabla^2\Gamma(\mathbf{x}) f_g(\mathbf{x}) + \nabla f_g(\mathbf{x}) \nabla\Gamma(\mathbf{x}). \quad (34)$$

Function $C(\mathbf{x})$ is assumed to be monotonic along the obstacle surface. This approach can fail with planar surfaces, where $\nabla^2\Gamma(\mathbf{x})$ is not full rank, or when $\nabla\Gamma(\mathbf{x})$ and $f_g(\mathbf{x})$ are collinear (i.e., requiring a saddle point correction identical to that in (19), as this represents a stationary point of $\nabla\Gamma(\mathbf{x}) \cdot f_g(\mathbf{x})$). In the latter case, this methodology can be easily combined with the approach discussed in Section V-A

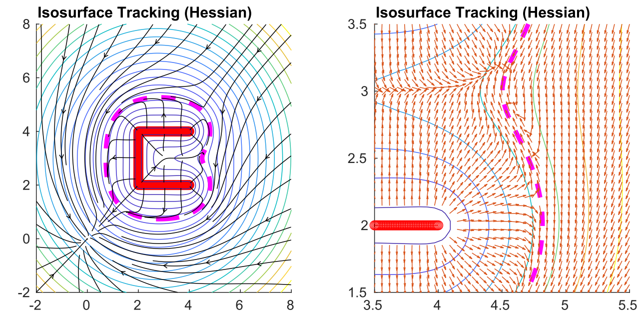
$$\phi(\mathbf{x}) = \nabla^2\Gamma(\mathbf{x}) f_g(\mathbf{x}) + \nabla f_g(\mathbf{x}) \nabla\Gamma(\mathbf{x}) + \delta \cdot \psi(\langle \mathbf{n}, \mathbf{v} \rangle). \quad (35)$$

The approach will also fail for nonconvex isosurfaces, where local maxima in $C(\mathbf{x})$ exist. Note also that this function can cause large velocities when $\Gamma(\mathbf{x})$ changes rapidly, as is the case for exponentials; as a result, this method requires normalization and consideration for numerical error. The approach is suitable for convex isosurfaces where $\det(\nabla^2\Gamma(\mathbf{x})) \neq 0 \forall \mathbf{x} \notin \mathcal{O}/\{\mathbf{x} : \langle \hat{\mathbf{n}}, \hat{\mathbf{v}} \rangle = 1\}$ (i.e., where there are no inflection points outside of those occurring when $\nabla\Gamma(\mathbf{x})$ and $f_g(\mathbf{x})$ are collinear, and the isosurface has some curvature).

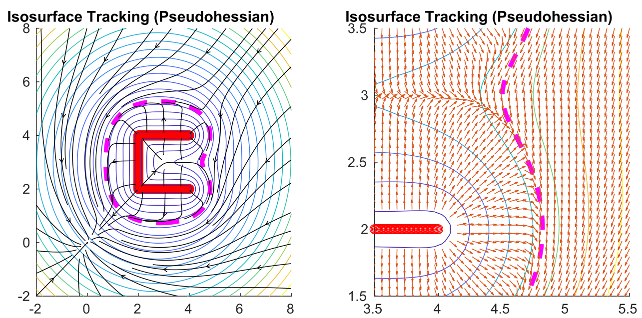
Proposition 2 (Hessian-based Obstacle Exit Strategy Convergence): Assume $\det(\nabla^2\Gamma(\mathbf{x})) \neq 0 \forall \mathbf{x} \notin \mathcal{O}$ and $|f_g(\mathbf{x}) \cdot \nabla\Gamma(\mathbf{x})| \neq \|f_g(\mathbf{x})\|\|\nabla\Gamma(\mathbf{x})\|$, then, $\mathbf{E}^T\nabla^2\Gamma(\mathbf{x}) f_g(\mathbf{x}) \neq \mathbf{0}$. If $\mathcal{O}$ is convex, then, the obstacle exit strategy preserves the convergence properties of $f_g(\mathbf{x})$.

Proof: Follows from (33). ∎

2) PseudoHessian/Spherical Approximation: While use of the Hessian can enable excellent navigation capabilities for convex obstacles, it is sensitive to local maxima in (33), which are present in the case of nonconvex obstacles (by definition when assuming $f_g(\mathbf{x})$ is linear). An extension based on the Hessian approach that can overcome such issues with local maxima involves a relaxation of the manifold to a convex form. For example, consider an approximating manifold described by $\Gamma(\mathbf{x}) = (\mathbf{x} - \mathbf{r})^T(\mathbf{x} - \mathbf{r})$ (i.e., a unit sphere). For this function, the Hessian is $\nabla^2\Gamma(\mathbf{x}) = 2 \cdot \mathbb{I}$. Using the Hessian of the spherical approximation of the isosurface can improve navigation performance on manifolds with small concave regions, as illustrated in Fig. 6(b). However, the approximating manifold must be close



(a)



(b)

Fig. 6. Obstacle exit strategies: In the context of nonconvex obstacles, using convex manifold approximations can overcome local minima in Hessian-based obstacle exit strategies. (a) Using the standard Hessian on nonconvex isosurfaces can lead to spurious attractors on the isosurface. (b) Using a pseudoHessian (assuming a spherical isosurface) on nonconvex isosurfaces can overcome spurious attractors on the isosurface.

in form to that of the original isosurface in order to be effective, as the resulting velocity from $\phi(\mathbf{x})$ cannot be within the nullspace of $\mathbf{E}$. This method is particularly effective when the selected isosurface is further away from the obstacle. Fig. 6(a) and (b) contrast the use of the Hessian and pseudoHessian approaches.

Proposition 3 (PseudoHessian-based Obstacle Exit Strategy Convergence): Assume that nonconvex isosurface $\Gamma(\mathbf{x}) = \gamma$ can be approximated by function $\Gamma'(\mathbf{x})$, where $\Gamma'(\mathbf{x})$ is a convex isosurface. If $\nabla^2\Gamma'(\mathbf{x}) f_g(\mathbf{x}) \notin \mathcal{N}(\mathbf{E}_\Gamma) \forall \mathbf{x} \in \{\mathbf{x} : \Gamma(\mathbf{x}) = \gamma\}$, the obstacle exit strategy preserves the convergence properties of $f_g(\mathbf{x})$.

Proof: Follows from Theorem 2. ■

3) Geodesic Approximation: A simple strategy for overcoming the limitations of the Hessian inspired approaches is to sample a straight, short line on the manifold over a set of predetermined directions and, then, select the direction that minimizes the potential along the path with regard to the attractor (i.e., the direction that minimizes $\int_{\mathbf{x}_0}^{\mathbf{x}_f} \|f_g(\mathbf{x})\| d\mathbf{x}$ along a straight line on the obstacle manifold). The geodesic approximation approach is illustrated in Fig. 7.

Let $\mathbf{u}_0$ be a candidate direction in $\mathbb{R}^{d-1}$, subject to $\mathbf{u}_0 \notin \mathcal{N}(\mathbf{E}(\mathbf{x}))$. We can, then, construct a line $\mathbf{X} = [\mathbf{x}_0 \quad \mathbf{x}_1 \quad \ldots \quad \mathbf{x}_f]$ along an isosurface with associated potential, $\mathbf{p}(\mathbf{X}, \mathbf{u})$, using the following, where α is a constant that regulates step size:

$$\mathbf{x}_{i+1} = \alpha \cdot \mathbf{E}(\mathbf{x}_i)\mathbf{E}(\mathbf{x}_i)^T \mathbf{u}_i + \mathbf{x}_i \tag{36}$$

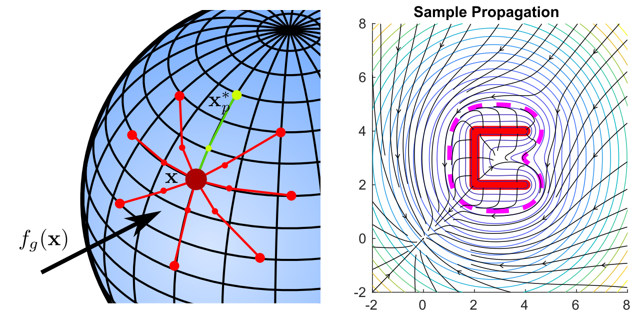


Fig. 7. Obstacle strategies: The geodesic approximation technique can be used in the context of nonconvex obstacles to seek the shortest path for circumventing an obstacle. The left figure depicts an illustration of the methodology, where samples are propagated in fixed directions at a fixed depth. The right figure shows a streamslice view of the geodesic approximation technique applied to a nonconvex obstacle.

$$\mathbf{u}_{i+1} = \frac{\mathbf{E}(\mathbf{x}_i)\mathbf{E}(\mathbf{x}_i)^T \mathbf{u}_i}{\|\mathbf{E}(\mathbf{x}_i)\mathbf{E}(\mathbf{x}_i)^T \mathbf{u}_i\|} \tag{37}$$

$$\mathbf{p}_{i+1} = \mathbf{p}_i + \alpha \cdot \|f_g(\mathbf{x}_{i+1})\|. \tag{38}$$

This method utilizes a first-order approximation to the obstacle surface; as such, it does not strictly find a path on the isosurface, but closely approximates one for appropriate α. The obstacle isosurface is a lower dimensional subspace of $\mathbb{R}^d$, which we treat as a locally Euclidean space in $\mathbb{R}^{d-1}$. For $\mathbb{R}^2$, the manifold is 1-D, allowing us to select $\mathbf{u}_0$ as ± 1; for $\mathbb{R}^3$ the tangent space of the manifold is 2-D, from which 16 directions are constructed[4] based on the basis vectors $\mathbf{e}_1$ and $\mathbf{e}_2$. For notational simplicity, we refer to the number of paths as M.

In the general application of this algorithm, the direction associated with the lowest potential is selected as input to $\phi(\mathbf{x})$. However, care must be taken with both the step size α, and number of steps N. If N or α are too large, wrapping or over extension can occur, leading to both unnecessary computation and inaccurate approximations, while values that are too small can lead to local minima. As such, search directions benefit from termination criteria, or careful calibration to ensure that both N and α are appropriate in the context of the obstacles they are being used to circumnavigate.

Our implementation of this algorithm includes an optional gradient descent phase, where the start position for sample calculation, $\mathbf{x}_0$, is projected onto the obstacle isosurface to be tracked (i.e., $\mathbf{x}'_0$ s.t. $\Gamma(\mathbf{x}'_0) = \gamma_c$). This can result in more consistent behavior for two reasons: First, the obstacle manifold maintains a consistent size; second, it maintains a consistent shape—a highly relevant fact when multiple obstacles are under consideration, as higher isosurfaces can encompass multiple obstacles, leading to inconsistent behavior of this method over changing isosurfaces.

The introduction of termination criteria can also provide a level of consistency and generalizability, while improving

[4]Sampling of these directions is a design consideration. In this work we use 16 linearly spaced search directions described by $\mathbf{u}^{(i)} = \sin(\pi/4 \cdot i)\mathbf{e}_1 + \cos(\pi/4 \cdot i)\mathbf{e}_2$. This method can be trivially extended to enable different sampling strategies.

computation times. For closed isosurfaces in 2-D, we can consider the termination criteria to be where $\nabla\Gamma(\mathbf{x}) \cdot f_g(\mathbf{x})$ is at a maximum or greater than a threshold and where $\|f_g(\mathbf{x})\| < \|f_g(\mathbf{x}_0)\|$[5]. For isosurfaces in 2-D that are not closed (e.g., $\Gamma(\mathbf{x}) = \exp(\|\mathrm{diag}([0 \quad 1])\mathbf{x}\|)$, representing the line $y = 0$), the criterion still applies in a single direction, resulting in lower potential in that direction.

For obstacles in $\mathbb{R}^d, d > 2$, it is not guaranteed that any set of candidate paths describing a straight line on the manifold will satisfy the termination criteria outlined above (i.e., the optimal path will only be guaranteed when the number of paths $M \to \infty$). To improve the chance of reaching the termination condition, we can bias search directions toward maximizing $f_g(\mathbf{x}) \cdot \nabla\Gamma(\mathbf{x})$ by including the derivative in the search direction when in proximity to a maximum. Note that this modifies the search direction update as follows:

$$\mathbf{u}_{i+1} = (1 - \sigma_z)\mathbf{u}_i + \sigma_z \nabla g(\mathbf{x}) \tag{39}$$

$$\sigma_z = \max(0, (\hat{g}(\mathbf{x}))^z) \tag{40}$$

$$g(\mathbf{x}) = \nabla\Gamma(\mathbf{x}) \cdot f_g(\mathbf{x}) \tag{41}$$

$$\nabla g(\mathbf{x}) = \nabla^2\Gamma(\mathbf{x}) \cdot f_g(\mathbf{x}) + \nabla f_g(\mathbf{x}) \cdot \nabla\Gamma(\mathbf{x}) \tag{42}$$

where $z \in \mathbb{R}^+$ is a scaling power that controls the effect (i.e., $z \to \infty \Rightarrow \sigma_z(g(\mathbf{x})) \to \delta(g(\mathbf{x}) - 1)$). While this method does not guarantee convergence for any obstacle, it can improve performance by biasing each sample to a point of termination at the expense of increased computation time.[6]

The termination criteria may fail for nonlinear $f_g(\mathbf{x})$. The application of geodesic approximation is discussed further in Section VII for a variety of complex scenarios. The advantage of this algorithm is that it runs in constant time (making it suitable for a real-time system), is easily parallelized for multiple search directions, and can provide near-optimal behavior with appropriate parameters.

Proposition 4 (Geodesic Approximation Exit Strategy Convergence): Let $\mathcal{M}$ be a closed, smooth manifold on $\{\Gamma(\mathbf{x}) = \gamma : \mathbf{x} \in \mathbb{R}^d\}$. Then, for $\alpha \ll 1, N \to \infty$, and $M \to \infty$, this obstacle exit strategy will converge to the path of minimum potential and preserve the convergence properties of $f_g(\mathbf{x})$.

Proof: Follows from $\alpha \ll 1$ and $\dot{\mathbf{x}}_{\mathcal{M}} \approx \mathbf{E}\mathbf{E}^T\mathbf{u}$. ■

All modulation strategies presented in this section assume a single function $\Gamma(\mathbf{x})$ that can represent the boundaries of arbitrarily shaped obstacles, under the further assumption that such functions follow (3)–(5). In the following section, we present functions that enable the arbitrary representation of complex shapes.

VI. Obstacle Representations

Ideal obstacle representations should allow for arbitrary representation of surfaces, paths, and boundaries. In this section, we introduce two obstacle representations that work toward this goal: a sample-based method and a projection-based method. The sample-based approach enables representation of arbitrarily complex obstacles, whereas the projection-based method allows efficient representation of mathematically simple surfaces, such as planes, cylinders, and ellipsoids. Fig. 2 depicts an example of the sample-based methodology, demonstrating navigation around nonconvex obstacles, as well as the applicability of the method to pointclouds. In this section, we provide explicit equations for the $\mathcal{C}^2$ differentiable obstacle representation functions, along with associated closed-form derivatives and Hessians.

[5]Concave or highly irregular isosurfaces may result in points on the obstacle manifold that maximize $\nabla\Gamma(\mathbf{x}) \cdot f_g(\mathbf{x})$, but where $\|f_g(\mathbf{x})\| > \|f_g(\mathbf{x}_0)\|$. This can induce limit cycles, breaking convergence guarantees.

[6]Hessian calculations for sample-based obstacles can incur significant computational expense, particularly, when the exponential field is used.

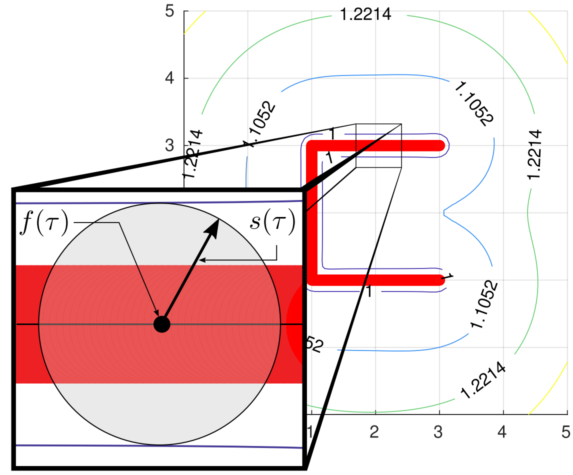


Fig. 8. Illustration of notation for sample-based obstacle representation: Here, $f(\tau)$ is a continuous function describing a path representation of the obstacle. The associated offset, $s(\tau)$, defines the width of the $\Gamma(\mathbf{x}) = 1$ boundary.

A. Sample-Based Obstacle Representation

Here, we introduce a sample-based obstacle representation for the purpose of representing complex, nonconvex surfaces. Consider a bounded region described by the continuous function $f(\tau) : \mathbb{R}^k \to \mathbb{R}^d, \tau \in [\tau_{\min}, \tau_{\max}]$, alongside boundary function $s(\tau) : \mathbb{R}^k \to \mathbb{R}$, which describes the boundary of the obstacle relative to $f(\tau)$, such that $\forall \mathbf{u} : \|\mathbf{u}\| = 1, \|f(\tau) + s(\tau) \cdot \mathbf{u}\| = 1 \Rightarrow \Gamma(\mathbf{x}) = 1$ (i.e., that $\Gamma(\mathbf{x}) = 1$ on the boundary). We introduce the *obstacle field*, $\Gamma_f(\mathbf{x}, \tau) : \mathbb{R}^d \times \mathbb{R} \to \mathbb{R}$ as a description of the sample point, and provide both a quadratic and exponential obstacle field, which can be used to induce different modulation effects by changing the rate at which the function $\Gamma(\mathbf{x})$ increases. The notation is illustrated in Fig. 8, whereas the effect of using either a quadratic or an exponential obstacle field can be seen in Fig. 10. The quadratic obstacle field can be constructed as follows:

$$\Gamma_f(\mathbf{x}, \tau) = \alpha \left(\|\mathbf{x} - f(\tau)\|^2 - s(\tau)^2 \right) + 1 \tag{43}$$

$$\nabla\Gamma_f(\mathbf{x}, \tau) = 2 \cdot \alpha \cdot (\mathbf{x} - f(\tau)) \tag{44}$$

$$\nabla^2\Gamma_f(\mathbf{x}, \tau) = 2 \cdot \alpha \cdot \mathbb{I}. \tag{45}$$

The exponential field can be constructed using the following equations. Note, that we introduce $f(\mathbf{x}) = \mathbf{x} - f(\tau), \|f(\mathbf{x})\|. = \|\mathbf{x} - f(\tau)\|$, and $\|f(\mathbf{x})\|_s = (\|\mathbf{x} - f(\tau)\| - s(\tau))$ to ease

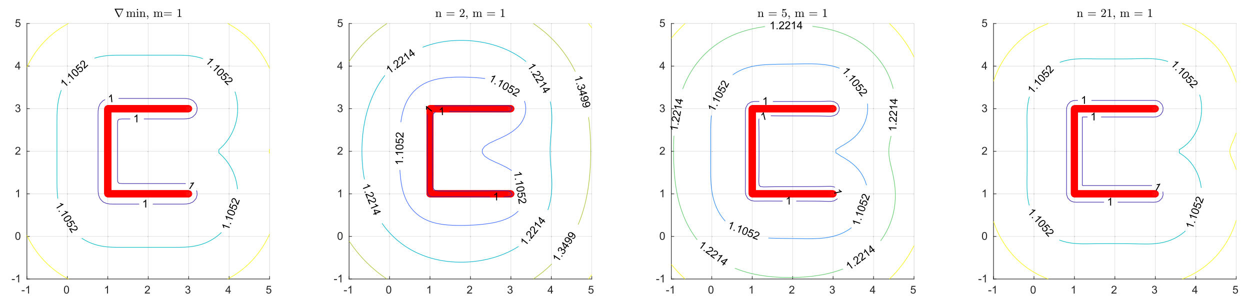


Fig. 9. Effect of increasing powers on minimum approximation. As the power of n in α_n increases, the function more closely approximates that produced by the differentiable minimum. The advantage of lower n-values is a smoother gradient, which leads to smoother modulation when navigating away from the obstacle.

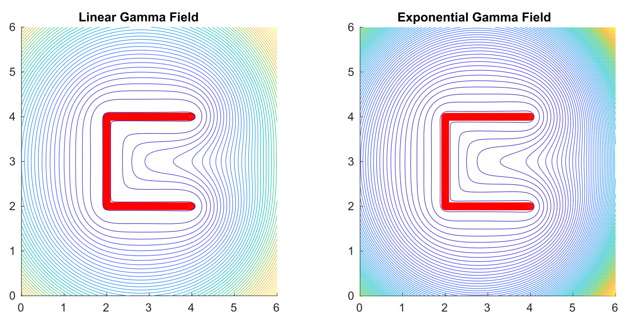


Fig. 10. Gamma field comparison: Quadratic and exponential gamma fields show similar levels when close to the obstacle boundary, but differ when far from the obstacle. Exponential fields are often a more appropriate choice in dense environments, enabling both a rapid onset of modulation (which can be considered to "kick-in" at $\Gamma(\mathbf{x}) \approx 30$ when using standard $\Lambda(\mathbf{x})$ matrices) and slow growth when close to the boundary.

notation

$$\Gamma_f(\mathbf{x},\tau) = \exp\left(\alpha \cdot \|f(\mathbf{x})\|_s^m\right) \tag{46}$$

$$\nabla\Gamma_f(\mathbf{x},\tau) = \alpha \cdot m \cdot \|f(\mathbf{x})\|_s^{m-1} \cdot \frac{f(\mathbf{x})}{\|f(\mathbf{x})\|_.} \cdot \Gamma_f(\mathbf{x},\tau) \tag{47}$$

$$\nabla^2\Gamma_f(\mathbf{x},\tau) = \alpha \cdot m \cdot \Gamma_f(\mathbf{x},\tau)\frac{\|f(\mathbf{x})\|_s^{m-2}}{\|f(\mathbf{x})\|_.^2} \left(\zeta \cdot f(\mathbf{x})f(\mathbf{x})^T + \|f(\mathbf{x})\|_s \|f(\mathbf{x})\|_. \cdot \mathbb{I}\right) \tag{48}$$

$$\zeta(\mathbf{x},\tau) = (m-1) + \alpha \cdot m \cdot \|f(\mathbf{x})\|_s \cdot \left(\alpha \cdot \|f(\mathbf{x})\|_. - s(\tau)\right)^{m-1} - \frac{\|f(\mathbf{x})\|_s}{\|f(\mathbf{x})\|_.} \tag{49}$$

Every point in obstacle field $f(\tau)$ exerts a degree of influence identical to that of a spherical obstacle, whereas the points in the field are infinitesimally close. The boundary of the field, where $\Gamma_f(\mathbf{x},\tau) = 1$, can be directly controlled through parameter $s(\tau)$, whereas field scaling can be controlled via the weighting parameter and the power to which the distance is raised (α and m, respectively).

The obstacle field must be integrated into a single obstacle representation that preserves the boundary of $\Gamma_f(\mathbf{x},\tau) = 1$. We use the following differentiable minimum function to construct $\Gamma_\nabla(\mathbf{x})$:

$$\Gamma_\nabla(\mathbf{x}) = -\frac{1}{\rho}\log\left(\int_\tau \exp\left(-\rho \cdot \Gamma_f(\mathbf{x},\tau)\right) d\tau\right). \tag{50}$$

Evaluating such an integral may yield an approach that is impractical for the user—particularly for more complex fields involving surfaces or sample data. Instead, we introduce a sampling-based approach to obstacle representation through $\hat{\Gamma}_\nabla(\mathbf{x})$ with a sample size of s. This function has the property that $\lim_{s\to\infty} \hat{\Gamma}_\nabla(\mathbf{x}) = \Gamma_\nabla(\mathbf{x})$ when uniform sampling is employed for τ

$$\hat{\Gamma}_\nabla(\mathbf{x}) = -\frac{1}{\rho}\log\left(\sum_\tau \exp\left(-\rho \cdot \Gamma_f(\mathbf{x},\tau)\right)\right) \tag{51}$$

$$\nabla\hat{\Gamma}_\nabla(\mathbf{x}) = \frac{\sum_\tau \exp\left(-\rho \cdot \Gamma_f(\mathbf{x},\tau)\right) \cdot \nabla\Gamma_f(\mathbf{x},\tau)}{\sum_\tau \exp\left(-\rho \cdot \Gamma_f(\mathbf{x},\tau)\right)} \tag{52}$$

$$\nabla^2\hat{\Gamma}_\nabla(\mathbf{x}) = \frac{1}{\sum w(\mathbf{x},\tau)}\left(\sum_\tau \zeta(\mathbf{x},\tau) - \left(\sum_\tau \nabla w(\mathbf{x},\tau)\right)\nabla\hat{\Gamma}_\nabla(\mathbf{x})^T\right) \tag{53}$$

$$\zeta(\tau,\mathbf{x}) = \left(\nabla w(\mathbf{x},\tau) \cdot \nabla\Gamma_f(\mathbf{x},\tau)^T + w(\mathbf{x},\tau) \cdot \nabla^2\Gamma_f(\mathbf{x},\tau)\right) \tag{54}$$

$$w(\mathbf{x},\tau) = \exp(-\rho \cdot \Gamma_f(\mathbf{x},\tau)) \tag{55}$$

$$\nabla w(\mathbf{x},\tau) = -w(\mathbf{x},\tau)\rho\nabla\Gamma_f(\mathbf{x},\tau). \tag{56}$$

While this approach allows statistical approximation to $\min_\tau \Gamma_f(\mathbf{x},\tau)$, $\hat{\Gamma}_\nabla(\mathbf{x})$ is sensitive to parameter ρ and can lead to nonconvex obstacle boundaries for a small number of samples, as well as sharp ridges in the gradient of $\Gamma(\mathbf{x})$. To smooth both the obstacle boundary and the gradient while removing dependence upon ρ, an approximation is introduced based on $\|\mathbf{x} - f(\tau)\|^n$; this resolves to a weighted approach for power n, defined as $\Gamma_n(\mathbf{x})$.

$$\Gamma_n(\mathbf{x}) = \frac{\sum_\tau \alpha_n(\mathbf{x},\tau)\Gamma_f(\mathbf{x},\tau)}{\sum_\tau \alpha_n(\mathbf{x},\tau)} = \frac{\rho(\mathbf{x})}{\beta(\mathbf{x})} \tag{57}$$

$$\nabla\Gamma_n(\mathbf{x}) = \frac{\beta(\mathbf{x})\nabla\rho(\mathbf{x}) - \rho(\mathbf{x})\nabla\beta(\mathbf{x})}{\beta(\mathbf{x})^2} \tag{58}$$

$$\nabla\beta(\mathbf{x}) = \sum_\tau \nabla\alpha_n(\mathbf{x},\tau) \tag{59}$$

$$\nabla\rho(\mathbf{x}) = \sum_\tau (\alpha_n(\mathbf{x},\tau)\nabla\Gamma_f(\mathbf{x},\tau) + \Gamma_f(\mathbf{x},\tau)\nabla\alpha_n(\mathbf{x},\tau)) \tag{60}$$

$$\nabla^2\Gamma_n(\mathbf{x}) = \frac{1}{\beta(\mathbf{x})}\left(\nabla^2\rho(\mathbf{x}) - \nabla\Gamma_n(\mathbf{x})\nabla\beta(\mathbf{x})^T - \nabla\beta(\mathbf{x})\nabla\Gamma_n(\mathbf{x})^T - \Gamma(\mathbf{x})\nabla^2\beta(\mathbf{x})\right) \tag{61}$$

$$\nabla^2\rho(\mathbf{x}) = \sum_\tau (\nabla\Gamma_f(\mathbf{x},\tau)\nabla\alpha_n(\mathbf{x},\tau)^T + \alpha_n(\mathbf{x},\tau) \cdot \nabla^2\Gamma_f(\mathbf{x},\tau) + \nabla\alpha_n(\mathbf{x},\tau)\nabla\Gamma_f(\mathbf{x},\tau)^T + \Gamma_f(\mathbf{x},\tau)\nabla^2\alpha_n(\mathbf{x},\tau)) \tag{62}$$

$$\nabla^2\beta(\mathbf{x}) = \sum_\tau \nabla^2\alpha(\mathbf{x},\tau). \tag{63}$$

Weights are based on powers of distances to the sample points of the obstacle, $f(\tau)$

$$\alpha_n(\mathbf{x},\tau) = \frac{1}{\|\mathbf{x}-f(\tau)\|^n} \tag{64}$$

$$\nabla\alpha_n(\mathbf{x},\tau) = \frac{-n(\mathbf{x}-f(\tau))}{\|\mathbf{x}-f(\tau)\|^{(n+2)}} \tag{65}$$

$$\nabla^2\alpha_n(\mathbf{x},\tau) = \frac{-n}{\|\mathbf{x}-f(\tau)\|^{(n+2)}}\left(\mathbb{I} - \frac{n+2}{\|\mathbf{x}-f(\tau)\|^2}(\mathbf{x}-f(\tau))(\mathbf{x}-f(\tau))^T\right). \tag{66}$$

The weights produce an effect analogous to the differentiable minimum as power increases. Fig. 9 illustrates this effect for various powers, demonstrating how the function converges to the differentiable minimum function as the power n increases. In practice, lower n-values ensure a smooth gradient stemming from the obstacle—a property leveraged in Section V to cause the system to generate motions that can escape from nonconvex regions when close to the obstacle boundary. Care should be taken to ensure that $s(\tau)$ and the sample size are chosen such that (5) remains true for $\Gamma(\mathbf{x}) > 1$ (i.e., that $\partial\mathcal{O} : \Gamma(\mathbf{x}) = 1$ is closed). The calculation of both $\Gamma(\mathbf{x})$ and $\nabla\Gamma(\mathbf{x})$ have a complexity of $O(s)$, where s represents the number of samples. In this work, the method operates without parallelization, with $s < 10\,\mathrm{k}$ to achieve real-time (< 1 ms) performance, ensuring the *computationally efficient* criteria discussed in Section III. However, this approach remains highly amenable to GPU acceleration, potentially allowing for much larger sample sizes; this representation is applicable to both complex surfaces and pointclouds.

B. Projection-based Obstacle Representation

While the representation introduced in Section VI-A enables arbitrary representation of complex obstacles, it can prove inefficient for representing large, relatively simple surfaces, such as the boundaries of a robot's reach or the surface upon which the robot operates. In this section, we introduce an obstacle representation using projections and normal equations.

Consider an obstacle representation based on projection matrix P, center $\mathbf{r}$, and offset s

$$\Gamma(\mathbf{x}) = \exp(\alpha\cdot(\|P(\mathbf{x}-\mathbf{r})\| - s)). \tag{67}$$

Then, the gradient and Hessian of this function can be shown as the following:

$$\nabla\Gamma(\mathbf{x}) = \alpha\cdot\frac{\Gamma(\mathbf{x})}{\|P(\mathbf{x}-\mathbf{r})\|}P^TP(\mathbf{x}-\mathbf{r}) \tag{68}$$

$$\nabla^2\Gamma(\mathbf{x}) = \alpha\cdot\frac{\Gamma(\mathbf{x})}{\|P(\mathbf{x}-\mathbf{r})\|}\left(P^TP + \left(\frac{\alpha}{\|P(\mathbf{x}-\mathbf{r})\|} - \frac{1}{\|P(\mathbf{x}-\mathbf{r})\|^2}\right)P^TP(\mathbf{x}-\mathbf{r})(\mathbf{x}-\mathbf{r})^TP^TP\right). \tag{69}$$

For a flat surface (e.g., a table or wall), the obstacle can be parameterized with normal $\mathbf{n}$ and $c(\mathbf{x}) = P\mathbf{c}$, where $\mathbf{c}$ is a point on the surface. The projection in such a case can be constructed as follows:

$$P = \frac{\mathbf{n}\mathbf{n}^T}{\mathbf{n}^T\mathbf{n}}. \tag{70}$$

A cylindrical obstacle representation is also easily represented for a given center $\mathbf{r}$, using a cylindrical projection P. For example, a vertical cylindrical projection function takes the following form:

$$P\mathbf{x} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 0\end{bmatrix}\mathbf{x} = \begin{bmatrix}x\\ y\\ 0\end{bmatrix}.$$

This format is also amenable to ellipsoidal objects, or for representing uncertainty about an obstacle. In the latter case, for obstacles with Gaussian uncertainty (as described by mean μ and covariance Σ), this can be accomplished using the Cholesky decomposition of the information matrix, $P = \mathrm{chol}(\Sigma^{-1})$, $c_{\text{dist}} = 1$ for the 1-σ bound, and $\mathbf{r} = \mu$. Used thusly, the boundary of the obstacle will be at the 1-σ bound. Fig. 11 depicts examples of both cylindrical and planar obstacles, whereas Fig. 12 shows an example of the covariance obstacle.

C. Multiple Obstacles

In contrast with prior approaches, that handled multiple obstacles by combining modulations, our methodology does so by combining their representations into a single function. This is accomplished using a differentiable minimum approach identical to that described in (51)–(56) but operating over obstacle representations $\Gamma_i(\mathbf{x})$

$$\hat{\Gamma}_\nabla(\mathbf{x}) = -\frac{1}{\rho}\log\left(\sum_i \exp\left(-\rho\cdot\Gamma_i(\mathbf{x})\right)\right) \tag{71}$$

$$\nabla\hat{\Gamma}_\nabla(\mathbf{x}) = \frac{\sum_i \exp\left(-\rho\cdot\Gamma_i(\mathbf{x})\right)\cdot\nabla\Gamma_i(\mathbf{x})}{\sum_i \exp\left(-\rho\cdot\Gamma_i(\mathbf{x})\right)} \tag{72}$$

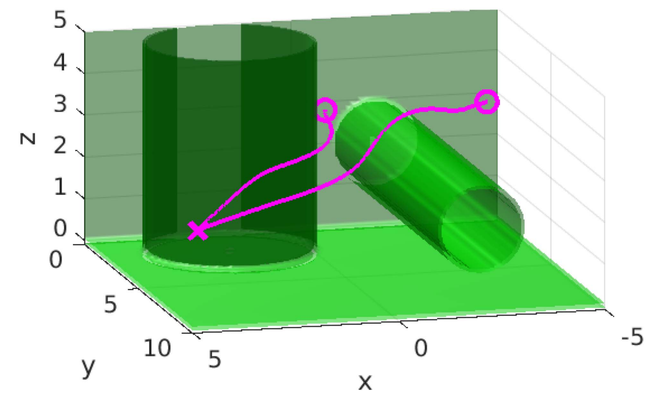


Fig. 11. Example of projection-based obstacles. The figure depicts two cylindrical obstacles and two planar obstacles, all in orthogonal orientations. The lines represent two examples of the robot successfully navigating around these obstacles. The simulations were performed using RK4, with $\Delta t = 1$ ms.

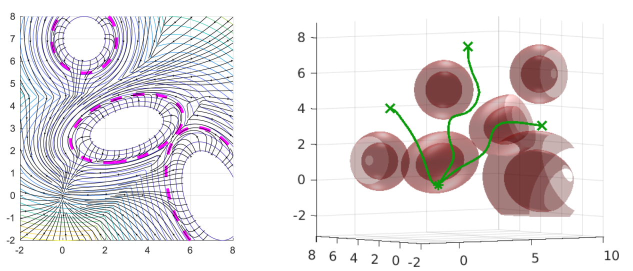


Fig. 12. Covariance obstacle in 2-D (left) and 3-D (right): The explicit representation of uncertainty ensures that modulated DS never penetrate the uncertainty region. A streamslice view is shown for the 2-D representation, whereas several RK4 simulations are depicted for the 3-D representation. Isosurfaces shown in 3-D are for $\Gamma(\mathbf{x}) = \{1, 1.4\}$

$$\nabla^2 \hat{\Gamma}_{\nabla}(\mathbf{x}) = \frac{1}{\sum w(\mathbf{x})} \left(\sum_i \zeta(i, \mathbf{x}) - \left(\sum_i \nabla w(\mathbf{x}) \right) \nabla \hat{\Gamma}_{\nabla}(\mathbf{x})^T \right) \tag{73}$$

$$\zeta(i, \mathbf{x}) = \left(\nabla w(\mathbf{x}) \cdot \nabla \Gamma_i(\mathbf{x})^T + w(\mathbf{x}) \cdot \nabla^2 \Gamma_i(\mathbf{x}) \right) \tag{74}$$

$$w(\mathbf{x}) = \exp(-\rho \cdot \Gamma_i(\mathbf{x})) \tag{75}$$

$$\nabla w(\mathbf{x}) = -w(\mathbf{x}) \rho \nabla \Gamma_i(\mathbf{x}). \tag{76}$$

Leveraging a single obstacle representation allows a single modulation approach to govern dynamics. This is preferable in the context of planning as the on-manifold dynamics operate over the isosurfaces generated by all obstacles, and require no further blending or optimization. Note that the differentiable minimum function used here can cause sharp ridges in the obstacle representation for large values of ρ, and combining representations in this way violates the monotonicity requirement stated in (5) (breaking requirements for convergence under standard modulation techniques). Using the on-manifold planning approach, monotonicity is not explicitly required for each and every point to ensure convergence; instead, monotonicity is only strictly required below the shifted boundary $\Gamma(\mathbf{x}) = \gamma$ for which this property is leveraged by the gradient ascent. Examples of modulation using multiple obstacles can be seen in Fig. 12, as well as within the detailed discussion in the following section.

VII. Evaluation

In this section, we evaluate our approach using a number of complex test cases for which existing methods are known to fail. We present two complex obstacle representations in $\mathbb{R}^2$, and two more in $\mathbb{R}^3$. For those in $\mathbb{R}^2$ we provide streamslice views alongside RK4 simulations demonstrating convergence through nonconvex obstacle regions; for those in $\mathbb{R}^3$, we present RK4 simulations only. We begin our discussion by demonstrating the contrast between our approach and the methods proposed by Zadeh and Billard [18] and Huber et al. [15]. We finalize our evaluation in a series of live experiments on a Franka Emika Panda Robot.

A. Simulated Examples

Fig. 13(a) demonstrates the use of our approach on a collection of 2-D projection-based spherical obstacles. For our approach, the obstacles are combined using the multiple obstacles interface, and the geodesic approximation method served as the obstacle exit strategy, with a termination condition of $\nabla\Gamma(\mathbf{x}) \cdot f_g(\mathbf{x}) / (\|\nabla\Gamma(\mathbf{x})\| \|f_g(\mathbf{x})\|) \geq 0.9$. The tracking level (depicted as the purple contour) was $\gamma = 1.6$, the tracking band gain $c_\gamma = 10$, and the geodesic approximation step size and max steps were 0.1, and 150, respectively. We contrast with the methods of Zadeh and Billard [18] and Huber et al. [15], demonstrating the failure of these algorithms to circumnavigate concave regions in this example. The additional blue crosses depicted in the approach by Huber et al. [15] represent the reference point placements.

Fig. 13(b) demonstrates 3-D circumnavigation of an explicitly concave obstacle (an uneven paraboloid, defined by $x = c \cdot (y^2 + z^2)$, $y \in [y_{\min}, y_{\max}]$, and $z \in [z_{\min}, z_{\max}]$), with the attractor on the opposing side of the concave region. Also shown are the results of 21 independent RK4 simulations when using the geodesic approximation method with step size and max steps of 0.05 and 150, respectively. Here, we also demonstrate how the existing methods by Zadeh and Billard [18] and Huber et al. [15] failed to circumnavigate this obstacle. Note that when outside the tracking region (i.e., $\Gamma(\mathbf{x}) \gg 1.7$), the simulations showed convergence to the original dynamics; however, as $\Gamma(\mathbf{x})$ converged to the tracking level, the geodesic approximation method identified efficient paths around the obstacle, leveraging the uneven surface and traveling along the y- and z-axes. In this example, the paraboloid was constructed using 2500 sample points. The step size for the RK4 simulations in all figures is 0.01.

We empirically demonstrate that no spurious attractors existed when our method was applied. The choice of tracking band gain, and the level tracked have an important effect on the system performance; the former affected where obstacle avoidance behavior overrode traditional modulation dynamics, whereas the latter played a significant role in the path taken

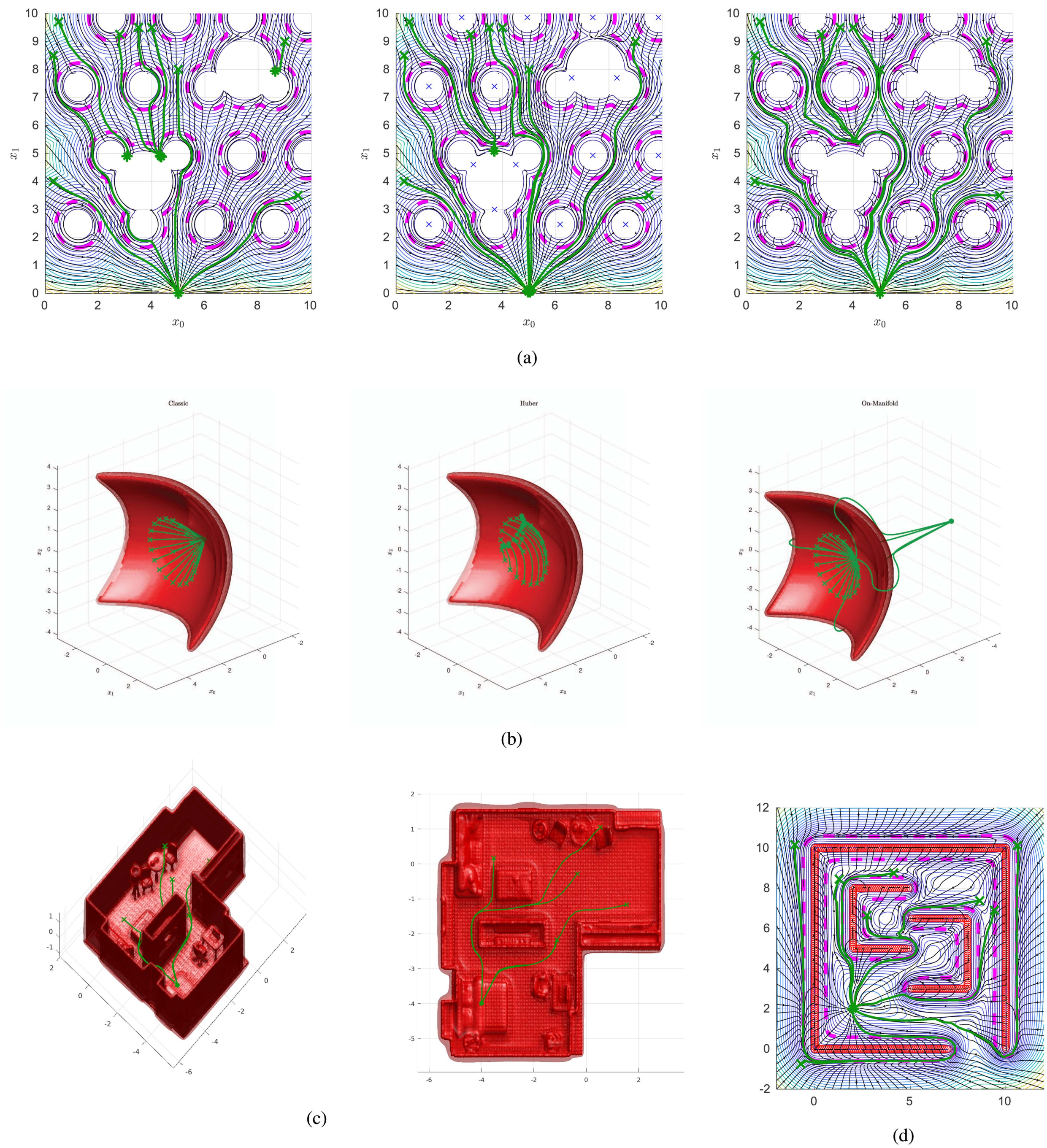


Fig. 13. Various examples: All on-manifold examples were generated using the geodesic approximation approach. (a) Circles example: A mixed convex and nonconvex obstacle space utilizing the projection based obstacle representation and multiple obstacle interface. Shown in each is the contour plot of the obstacle representation, $\Gamma(\mathbf{x})$, overlaid with the streamslice view of the modulated system as well as several RK4 simulations from various positions around the obstacles (beginning at green cross positions and terminating at green star positions). From left to right, we depict the "classic," "Huber," and "on-manifold" approaches (ours). Only the "on-manifold" approach demonstrates global convergence. (b) Parabolic example: Demonstration of 3-D navigation behavior in the presence of an explicitly concave obstacle, alongside RK4 simulations initialized in the concave region of the obstacle (beginning at green cross positions and terminating at green star positions). From left to right, we depict the "classic," "Huber," and "on-manifold" approaches (ours). Only the "on-manifold" approach demonstrates global convergence. (c) Flat example: Demonstration of 3-D navigation behavior in a home environment. Shown is an isometric view (left) and a top down view (right). This example was generated using the geodesic approximation approach. Dataset source: Schmid et al. [34]. (d) Gated environment example: A nonconvex obstacle space utilizing the sample-based obstacle representation.

around nonconvex obstacles before traditional dynamics could again be utilized. Particular care must be taken with the tracking level, as setting this value too high can lead to local minima in the gradient ascent portion of the algorithm—or, in extreme cases, portions of the state-space, which the robot cannot enter or exit. In general, isosurface selection can be intuitively selected based on the problem at hand.

In Fig. 13(c) we demonstrate navigation within a cluttered 3-D environment, obtained from a simulated point cloud representation of a European apartment [34]. The dataset was

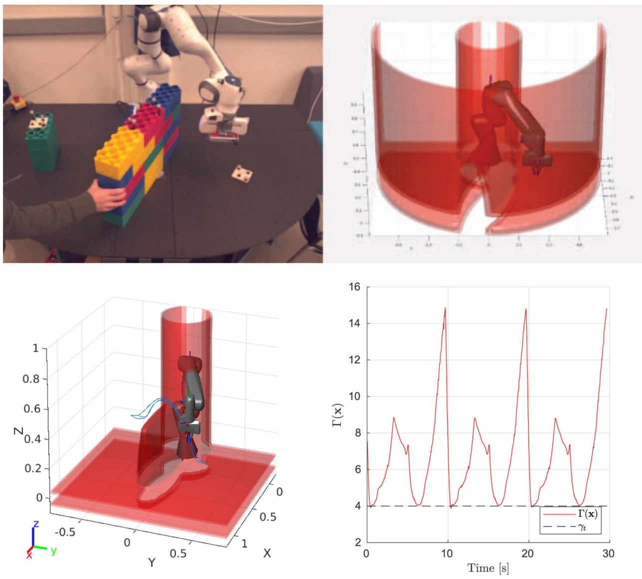


Fig. 14. Top: Image of a robot demonstration, alongside a visual representation of the robot and associated isosurfaces (see text). Bottom left: Path taken by the robot to avoid the wall obstacle (with the outer wall removed). Bottom right: Graph of the obstacle representation function, with a tracking value of $\gamma_t = 4$. Note, the robot did not enter the regions of $\Gamma(\mathbf{x}) \leq \gamma_t$, demonstrating impenetrability.

downsampled to 19 100 points, and the reconstruction was of the isosurfaces of $\Gamma(\mathbf{x}) = 1.1$ (depicted with a transparency of 0.8), and $\Gamma(\mathbf{x}) = 1.4$ (depicted with a transparency of 0.3). Note, how the larger isosurface experienced a greater level of smoothing, which is analogous to the smoother contours observed in the 2-D examples. The tracking level used in this example is $\Gamma(\mathbf{x}) = 1.6$. The figure shows several RK4 simulations, demonstrating efficient circumnavigation of the obstacles in the scene. Similarly, Fig. 13(d) showcases the updated modulation strategy acting in a nonconvex 2-D space where a large gated box envelops two smaller C-shaped obstacles. The obstacles were represented in a single sample-based obstacle representation, and the parameters used were identical to those in the circle environment.

Fig. 13(a) and (b) explicitly demonstrate the benefits of our methodology over existing methods. Fig. 13(c) and (d) demonstrate the utility of the sample-based representation, where complex scenes can be efficiently modeled and circumnavigated using our approach.

B. Robot Implementation

We applied the modulation approach to real-time collision avoidance using a Franka Emika Panda robot, as illustrated in Fig. 14. The system operated using a 1 KHz real-time control loop, incorporating 100 Hz position updates of obstacles in the scene as obtained from an Optitrack motion capture system. The modulation was calculated and applied at each iteration of the control loop, whereas the geodesic approximation algorithm was off-loaded to a separate thread and ran dynamically based on the problem at hand (typically at 50–100 Hz) whereas the previous result was propagated with each control cycle (i.e., $\mathbf{u}_{i+1} = \mathbf{E}(\mathbf{x})\mathbf{E}(\mathbf{x})^T\mathbf{u}_i$). When state information was received from the Optitrack motion capture system, the system updated the positions of targets and obstacles in the scene between control cycles. Modulations were applied to the states of the robot end-effector, guiding the robot between two targets tracked via the motion capture system. The orientation of the end-effector was not affected by the dynamical system in the experimental setup. The targets could be moved during operation, demonstrating the real-time nature of the system. The dynamical system produced Cartesian velocity commands that were translated to robot torques using a custom velocity controller, which is based on a joint impedance controller but modified with additional compensating controllers that bias the orientation of the robot to a vertical orientation. For safety, the system velocity was constrained to 0.35 m/s, and leverages torque feedback to immediately stop upon impact with any obstacle in the scene.

The system was implemented in C++ without GPU acceleration. Several threads did operate on the system, acting asynchronously to: calculate the robot control loop (modulation, target tracking, etc.), update obstacle positions, and record state information. The system did not make use of an external communication framework, such as ROS [29], or LCM [14], but instead leveraged shared memory to minimize communication overhead and ensured the system had high bandwidth and remained responsive. We implemented the experimental system on Ubuntu 20.04 with a real-time Linux kernel, using a computer containing an Intel i9-9900 k (5 GHz) and 32 GB DDR4-3200 MHz RAM.

The general obstacle representation consists of a planar surface representing the table on which the robot is mounted ($\mathbf{n} = \{0, 0, 1\}, \mathbf{c} = \mathbf{0}$); a small, cylindrical obstacle representing the robot base at the center of the table ($\mathbf{r} = \mathbf{0}$, $c_{\text{dist}} = 0.2$ m), and a second cylindrical obstacle representing the reach of the robot ($\mathbf{r} = \mathbf{0}$, $c_{\text{dist}} = 0.9$ m). Note that in the calculation of all obstacles, the offset is based on half of the end effector width (i.e., a Minkowski sum).

We present the results from two experiments for evaluation. First is the circumnavigation of a moveable wall in the robot workspace, whereas the second experiment introduces a static nonconvex shape. In the first experiment, we observed successful robot circumnavigation of the obstacle at varying orientations. Fig. 14 showcases the robot's behavior for a wall in the robot workspace, alongside images of the robot setup, the path taken by the robot, and the obstacle representation function during the experiment. As can be seen from the obstacle representation function (i.e., $\Gamma(\mathbf{x})$), the robot never entered space in which $\Gamma(\mathbf{x}) < \gamma_t$.

In the second experiment, shown in Fig. 15, we observed robot circumnavigation of an explicitly nonconvex obstacle. Here, we once again observed the impenetrability properties of the modulation approach, and observed successful escape from nonconvex regions of the obstacle representation. Note that the modulation approach did not take into account the robot joints, and so this obstacle is projected further back into the workspace from its physical location.

In addition, we introduced a human obstacle composed of a representation of the human arm, and implemented as a line

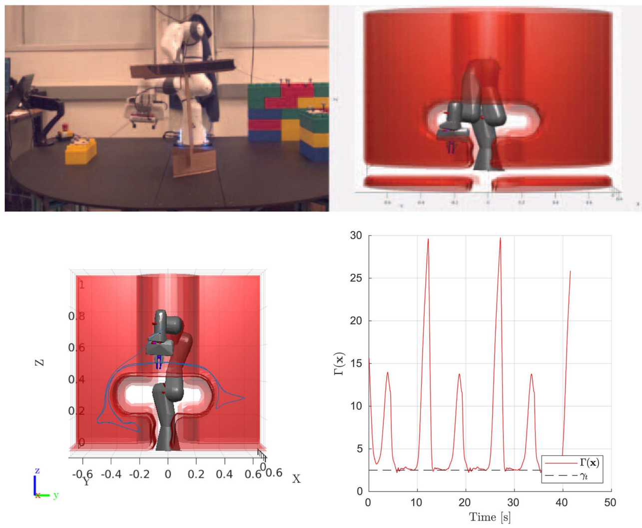


Fig. 15. Top: Image of a robot demonstration alongside a visual representation of robot and associated isosurfaces (see text). Bottom left: Path taken by the robot to avoid the T-shaped obstacle (with the outer wall removed). Bottom right: Graph of the obstacle representation function, with a tracking value of $\gamma_t = 2.5$. Note, the robot does not enter regions of $\Gamma(\mathbf{x}) \leq \gamma_t$, which demonstrates impenetrability.

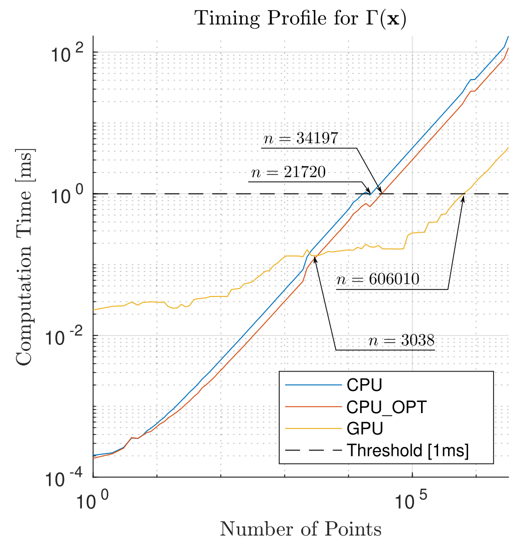


Fig. 16. Timing results.

between two markers mounted on the wrist and on the bicep close to the elbow, with points sampled at a spacing of 0.5 cm apart (typically resulting in ~ 80 points) and an obstacle radius of $s(\tau) = 10\,\text{cm}$. In these experiments, we introduced a velocity scaling that normalizes the modulation and applied a global scaling of $\max(0, 1 - 1/\Gamma(\mathbf{x}))$. This forces stopping behavior should the robot penetrate the obstacle boundary (such as if the human moves quickly into the space). See the accompanying video material for a demonstration of the robot operating in the presence of humans in the same workspace.

VIII. Discussion

A. Scope of Application

The methodology we present here has application to fast and efficient modeling of various environmental representations, including maps based on occupancy grids, simultaneous localization and mapping, or full geometric representations (e.g., an STL file, where appropriately dense vertex information can be utilized to represent the obstacle). The methodology enables the inclusion of a host of sensors, including RGBD sensors, laser scanners, or motion capture, while ensuring high system bandwidth (limited primarily by that of the sensor).

The methodology also has applications for human–robot interaction, where robot motions can be adapted to avoid collision or minimize impact with human collaborators [22]. DS are powerful tools in this context, as they help to ensure safety though a high bandwidth system. In combination with the velocity scaling we introduced in Section VII-B, our methodology ensures the robot slows to a halt when a human is nearby based on the boundary of $s(\tau)$; however, the dynamical system, and its associated bandwidth, are only as good as that of the sensing technology employed.

B. Performance

Our sample-based representation has $O(n)$ complexity with regard to the number of samples, whereas the modulation has $O(d^3)$ complexity, where d is the dimension of the data (assuming naïve multiplication). In order to evaluate runtime performance, we tested the sample-based algorithm with up to 10^6 points in $\mathbb{R}^3$. We also provide an optimized CPU implementation (using four-way loop unrolling), as well as a GPU implementation for comparison. The results are shown in Fig. 16, where `CPU_OPT` refers to the optimized CPU implementation.

For evaluation, we considered how many points could be utilized in the sample-based representation before a call to $\Gamma(\mathbf{x})$ reached 1 ms. The results in Fig. 16 demonstrated that the CPU implementation could reach between 20 k–35 k points depending on the optimization employed, whereas the GPU could consistently handle up to 600 k points while providing a result in under 1 ms (reported times are averages across 1000 samples). Note that GPU implementation only provides a tangible improvement over the CPU when using greater than 3 k points. When derivative information is also included in the calculation (see Fig. 17), we observed that the CPU could consistently handle up to 4 k points when using the exponential representation (up to 8 k for the quadratic representation), whereas the GPU could handle up to 155 k points (and up to 244 k points when using the quadratic representation). These runtimes were computed on Ubuntu 20.04 without a real-time kernel (Amd Ryzen 9 5950x, 128 GB DDR4 RAM, NVIDIA RTX 3090).

The results indicated a clear runtime performance benefit to utilizing a GPU over a CPU implementation for a large number of points, and a clear performance benefit to using the quadratic representation over the exponential representation (an approximately twofold speedup). Note that calculating the gradient information is required for the modulation.

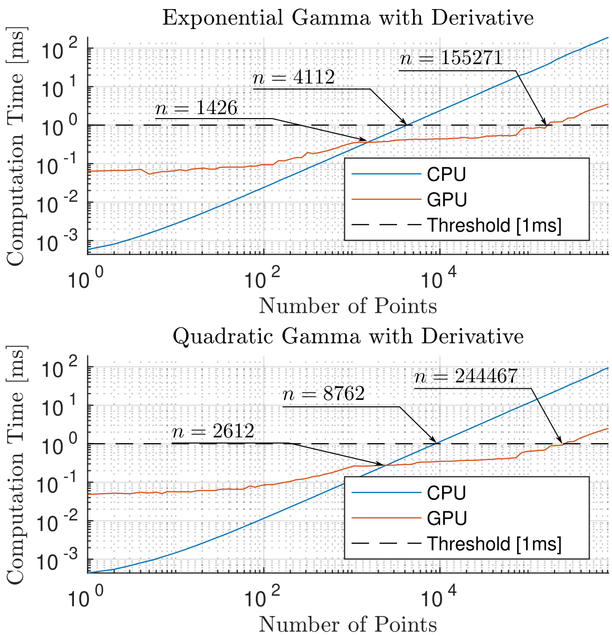


Fig. 17. Timing results with derivative.

C. Limitations

One primary limitation of this work is that we assume a Euclidean space in $\mathbb{R}^d$, and a fully holonomic dynamical system operating in $\mathbb{R}^d$ (i.e., the system operates over linear states, not rotations, and care should be taken to ensure states are universally reachable). This has implications for implementation as DS states should match the linear degrees of freedom of the target system, and boundary constraints (such as robot reach) should be treated as obstacles.

D. Scope for Further Research

As noted in Section VI-B, the projection and sample-based obstacle representations could be combined into a tesselated obstacle representation. This would offer a number of potential advantages, particularly the construction of exact meshes with downsampled data, leading to performance improvements and direct compatibility with existing computer graphics models. Second, while the geodesic approximation method demonstrates the effectiveness of such a solution, the approach could be improved in a number of ways—in particular, the approach could benefit from sampling heuristics to better improve geodesic approximation and remove small jerks associated with discretization. Finally, as noted above, the modulation methodology does not consider the full joint space of a robot manipulator. Extension of the modulation method to take the full kinematic chain of the robot (such as by Bylard et al. [6]) or an extension to nonEuclidean spaces could yield significant improvements to robustness.

IX. Conclusion

In this article, we introduced a novel DS planning framework based on on-manifold navigation that includes two approaches to obstacle representation. The sample-based obstacle representation enables representation of complex, nonconvex obstacles using a dense collection of sample points, whereas the projection-based obstacle representation enables computationally efficient representation of a number of common surfaces, such as tables and cylinders, as well as the representation of uncertainty. We successfully applied the modulation framework on a Franka Emika Panda robot in a human collision avoidance system, operating at 1 kHz, and provided several examples demonstrating obstacle circumnavigation for nonconvex obstacles in both 2-D and 3-D. We empirically demonstrate convergence for nonconvex obstacles provided that:

1) the obstacle representation value at the target position is higher than the chosen tracking level ($\Gamma(\mathbf{x}_f) > \gamma$);
2) the monotonicity requirements for multiple obstacles is satisfied (see Section VI-C); and
3) a path exists from the current position to the target position that does not cross the boundary associated with the tracking level.

Care should be taken when setting the tracking level to ensure that appropriate behavior results.

We believe the methodology presented here can enable efficient, realtime navigation around dynamic obstacles in a changing environment. Reactive navigation methods are required for environments with rapidly changing states such as close proximity human–robot interaction, or high-speed navigation in dense environments. Our method works to enable these capabilities by providing robust, reactive, high-bandwidth, and real-time navigation.

Appendix A
Impenetrability of Obstacle Boundary

According to the von Neuman boundary condition, impenetrability is ensured if the normal velocity at boundary points $\mathbf{x} \in \partial\mathcal{O}$ vanishes, i.e., $\mathbf{n}(\mathbf{x})^T\dot{\mathbf{x}} = 0\,\forall\mathbf{x} \in \partial\mathcal{O}$. Let us now consider (18), restated here for convenience

$$\dot{\mathbf{x}} = \underbrace{M(\mathbf{x})f_g(\mathbf{x})}_{\text{Classic Modulation}} + \underbrace{\mathbf{E}\cdot\phi(\mathbf{x})\mathbf{n}^T f_g(\mathbf{x})}_{\text{Tangent Space Dynamics}}.$$

The classic modulation approach guarantees impenetrability following the proof in [18, Th. 2] for convex obstacles and in [15, Th. 1] for arbitrary continuously differentiable obstacle boundaries as defined in (3)–(5). In short, since $\mathbf{n}(\mathbf{x})$ is equal to the first eigenvector of the orthogonal basis used to construct $M(\mathbf{x}) = H(\mathbf{x})^T\Lambda(\mathbf{x})H(\mathbf{x})$ and the first eigenvalue is zero$\forall\mathbf{x} \in \partial\mathcal{O}$, then, the normal velocity vanishes at the boundary for $M(\mathbf{x})f_g(\mathbf{x})$.

Given that the tangent space dynamics, by construction, are constrained to $\mathbf{E}$ due to the definition of the orthogonal basis $H(\mathbf{x})$; $\mathbf{n}(\mathbf{x}) = \nabla\Gamma(\mathbf{x}) \in \mathcal{N}(\mathbf{E})$, then, any velocity components generated by the tangent space dynamics will never induce velocities normal to the boundary. Hence, impenetrability guaranteed by the classic modulation term is preserved. ■

Appendix B
Convergence for Convex Obstacles

Following [11] and [3] we prove global stability and convergence of our modulation strategy using a Lyapunov function of the form $V(\mathbf{x}) = (\mathbf{x}-\mathbf{x}^*)^T\mathbf{P}(\mathbf{x}-\mathbf{x}^*)$ with $\nabla V(\mathbf{x}) = \mathbf{P}(\mathbf{x}-\mathbf{x}^*)$ and positive definite matrix $\mathbf{P} = \mathbf{P}^T \succ 0$. Global stability and convergence of a dynamical system can be ensured if the following Lyapunov stability conditions hold, $\dot{V}(\mathbf{x}) < 0$ $\forall \mathbf{x} \neq \mathbf{x}^*$, with $V(\mathbf{x}) > 0 \forall \mathbf{x} \neq \mathbf{x}^*$. For ease of readability let $\bar{\mathbf{x}} = (\mathbf{x}-\mathbf{x}^*)$. Following the work of Huber et al. [15], for $\dot{\mathbf{x}} = M'(\mathbf{x})f_g(\mathbf{x}), f_g(\mathbf{x}) = -(\mathbf{x}-\mathbf{x}^*)$, the time derivative of the Lyapunov function $\dot{V}(\mathbf{x})$ yields

$$\begin{aligned}\dot{V}(\mathbf{x}) &= -\bar{\mathbf{x}}^T\mathbf{P}M'(\mathbf{x})\bar{\mathbf{x}} - \bar{\mathbf{x}}^T M'(\mathbf{x})\mathbf{P}\bar{\mathbf{x}}\\ &= -\underbrace{2\cdot\bar{\mathbf{x}}^T\mathbf{P}M(\mathbf{x})\bar{\mathbf{x}}}_{\text{Classic Modulation}} - \underbrace{2\cdot\bar{\mathbf{x}}^T\mathbf{P}\mathbf{E}\mathbf{E}^T\phi(\mathbf{x})\mathbf{n}^T\bar{\mathbf{x}}}_{\text{Tangent Space Dynamics}} \qquad (77)\\ &= 2\cdot\dot{V}_C(\mathbf{x}) + 2\cdot\dot{V}_T(\mathbf{x}). \qquad (78)\end{aligned}$$

Note that by construction $M(\mathbf{x}) = M(\mathbf{x})^T$. Using (14), $\dot{V}_C(\mathbf{x})$ can be expressed as follows:

$$\begin{aligned}\dot{V}_C(\mathbf{x}) &= -\bar{\mathbf{x}}^T\mathbf{P}\left[\mathbb{I} + \frac{\mathbb{I} - 2\cdot\hat{\mathbf{n}}\hat{\mathbf{n}}^T}{\Gamma(\mathbf{x})}\right]\bar{\mathbf{x}} \qquad (79)\\ &= -\left(1+\frac{1}{\Gamma(\mathbf{x})}\right)\bar{\mathbf{x}}^T\mathbf{P}\bar{\mathbf{x}} + \frac{2}{\Gamma(\mathbf{x})}\bar{\mathbf{x}}^T\mathbf{P}\hat{\mathbf{n}}\hat{\mathbf{n}}^T\bar{\mathbf{x}} \leq 0. \qquad (80)\end{aligned}$$

When $\Gamma(\mathbf{x}) > 1$, $\dot{V}_C(\mathbf{x}) < 0 \forall \mathbf{x} \neq \alpha\cdot\hat{\mathbf{n}}, \alpha > 0$, as

$$\frac{2}{\Gamma(\mathbf{x})}\bar{\mathbf{x}}^T\mathbf{P}\hat{\mathbf{n}}\hat{\mathbf{n}}^T\bar{\mathbf{x}} < \left(1+\frac{1}{\Gamma(\mathbf{x})}\right)\bar{\mathbf{x}}^T\mathbf{P}\bar{\mathbf{x}}. \qquad (81)$$

Alternatively, when $\Gamma(\mathbf{x}) = 1$, then, $\dot{V}_C(\mathbf{x}) = 0 \iff \bar{\mathbf{x}} = \alpha\mathbf{n}, \Gamma(\mathbf{x}) = 1$; i.e., the spurious attractors on the obstacle boundary. Hence, $\dot{V}_C(\mathbf{x}) \leq 0 \,\forall \mathbf{x} \neq \mathbf{x}^*$.

Following (19) for $\phi(\mathbf{x})$ the tangent space dynamics are only activated when $\langle \mathbf{e}_i, \mathbf{v}\rangle \geq 0$ else $\dot{V}_T(\mathbf{x}) = 0$. Regarding negativity of $\dot{V}_T(\mathbf{x})$, assume that $\psi_k(\zeta) \to \delta(\zeta - 1)$ (explicit discontinuity), with $\bar{\mathbf{x}} = \alpha\mathbf{n}, \Gamma(\mathbf{x}) = 1$, then

$$\begin{aligned}\dot{V}_T(\mathbf{x}) &= -\bar{\mathbf{x}}^T\mathbf{P}\mathbf{E}\mathbf{E}^T\underbrace{\frac{c\cdot\delta(\langle\mathbf{n},\bar{\mathbf{x}}\rangle - 1)}{\Gamma(\mathbf{x})^2}}_{\phi(\mathbf{x}) = (19)}\cdot\mathbf{e}_i\mathbf{n}^T\bar{\mathbf{x}} \qquad (82)\\ &= -\frac{c\cdot\alpha^2}{\Gamma(\mathbf{x})^2}\mathbf{n}^T\mathbf{P}\mathbf{e}_i < 0. \qquad (83)\end{aligned}$$

Since $\mathbf{P} \succ 0$, then, $\mathbf{n}^T\mathbf{P}\mathbf{e}_i > 0$. Hence, $\dot{V}(\mathbf{x}) < 0 \forall \mathbf{x} \neq \mathbf{x}^*$, proving that convergence to $\mathbf{x}^*$ is guaranteed. ∎

Appendix C
Convergence of (26)

Consider point $\mathbf{x}' \in \{\mathbf{x} : \Gamma(\mathbf{x}) = \gamma\}$, s.t. $\hat{\mathbf{v}}\cdot\hat{\mathbf{n}} = 1, \mathbf{v} = f_g(\mathbf{x}')$, and $\mathbf{n} = \nabla\Gamma(\mathbf{x}')$. At this point, $\dot{\mathbf{x}}' = M(\mathbf{x})f_g(\mathbf{x}') = \alpha f_g(\mathbf{x}')$, where $\alpha \geq \epsilon/\Gamma(\mathbf{x})^2 > 0$. If $\phi(\mathbf{x})$ presents a consistent path to $\mathbf{x}'$, then, circumnagivation of the obstacle is guaranteed. For linear DS, $f_g(\mathbf{x})$, this ensures convergence to the obstacle from the manifold, provided that $\Gamma(\mathbf{x}_f) > \gamma, \mathbf{x}_f = \lim_{t\to\infty}\int_0^t f_g(\mathbf{x})dt + \mathbf{x}_0$. Note that outside the shifted boundary, the modulation is identical to that of Zadeh and Billard [18] with a shifted boundary, and similarly inherits the impenetrability and convergence conditions (see Proposition 1).

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